

Recent advances in graphene-based phase change composites for thermal energy storage and management

Qiang Zhu^{a,b,c,1}, Pin Jin Ong^{a,1}, Si Hui Angela Goh^a, Reuben J. Yeo^a, Suxi Wang^a,
Zhiyuan Liu^{d,**}, Xian Jun Loh^{a,b,e,*}

^a Institute of Materials Research and Engineering, A*STAR (Agency for Science, Technology and Research), 2 Fusionopolis Way, Innovis, #08-03, 138634, Singapore

^b Institute of Sustainability for Chemicals, Energy and Environment, A*STAR (Agency for Science, Technology and Research), 1 Pesek Road, Jurong Island, 627833, Singapore

^c School of Chemistry, Chemical Engineering and Biotechnology, Nanyang Technological University, 21 Nanyang Link, 637371, Singapore

^d Shenzhen Institutes of Advanced Technology (SIAT), Chinese Academy of Sciences (CAS), 1068 Xueyuan Avenue, Shenzhen, 518055, China

^e Department of Material Science and Engineering, National University of Singapore, 9 Engineering Drive 1, #03-09 EA, 117575, Singapore

ARTICLE INFO

Keywords:

Phase change material
Nanocomposites
Solar energy
Sustainable energy
Thermo-regulation

ABSTRACT

Energy storage and conservation are receiving increased attention due to rising global energy demands. Therefore, the development of energy storage materials is crucial. Thermal energy storage (TES) systems based on phase change materials (PCMs) have increased in prominence over the past two decades, not only because of their outstanding heat storage capacities but also their superior thermal energy regulation capability. However, issues such as leakage and low thermal conductivity limit their applicability in a variety of settings. Carbon-based materials such as graphene and its derivatives can be utilized to surmount these obstacles. This study examines the recent advancements in graphene-based phase change composites (PCCs), where graphene-based nanostructures such as graphene, graphene oxide (GO), functionalized graphene/GO, and graphene aerogel (GA) are incorporated into PCMs to substantially enhance their shape stability and thermal conductivity that could be translated to better storage capacity, durability, and temperature response, thus boosting their attractiveness for TES systems. In addition, the applications of these graphene-based PCCs in various TES disciplines, such as energy conservation in buildings, solar utilization, and battery thermal management, are discussed and summarized.

1. Introduction

In recent decades, the escalating worldwide energy consumption, in conjunction with the rapid depletion of non-renewable energy sources and the amplification of greenhouse gas emissions, has made it necessary for sustainable and environmentally friendly energy alternatives to be explored [1–5]. Renewable energy sources such as solar and wind power are considered inexhaustible; however, their availability is contingent upon the presence of wind, cloud cover and daylight hours for solar energy collection, and are therefore subject to periodic fluctuations. Energy storage has the potential to address significant energy fluctuations and enhance energy utilization while mitigating carbon emissions by temporarily storing and releasing energy under specific conditions. In

this way, a constant energy output can be available, on-demand, at all times of the day and year. Nevertheless, with the ever-increasing global demand for energy, there has been a significant surge in global interest in making energy storage technologies more efficient in contemporary times, in terms of storage capacity, charging and discharging times, battery lifetime, etc. Hence, the advancement of materials for energy storage is crucial.

Extensive research has been carried out pertaining to phase change materials (PCMs) in this aspect. PCMs are a class of materials that have the ability to absorb and release large amounts of thermal energy when they undergo a phase change. The phase transition temperature is one of the core characteristics of these materials that determine the type of application that it can be used in. Currently, PCMs have been broadly

* Corresponding author. Institute of Materials Research and Engineering, A*STAR (Agency for Science, Technology and Research), 2 Fusionopolis Way, Innovis, #08-03, 138634, Singapore.

** Corresponding author.

E-mail address: lohxj@imre.a-star.edu.sg (X.J. Loh).

¹ These authors contribute equally.

applied in buildings [6,7], solar energy utilization [8–10], thermoelectric materials [11–14], textiles [15,16], and electronics [17,18]. PCMs are divided into four categories: solid-solid PCMs (SSPCMs), solid-liquid PCMs (SLPCMs), solid-gas PCMs (SGPCMs), and liquid-gas PCMs (LGPCMs) [19,20]. Although SGPCMs and LGPCMs have higher latent heat storage capacities, the large volumetric changes during phase transition limit their use in commercial applications. SSPCMs, on the other hand, experience small volume changes but they have a small latent heat capacity and have only recently been explored for a few potential future applications [21,22]. Therefore, SLPCMs are one of the most widely used PCMs due to their high latent heat storage capacity, small volume change, and good chemical and thermal stability [23,24].

SLPCMs can be mainly classified into organic and inorganic PCMs. Inorganic PCMs include crystalline hydrated salts and metals and they have high energy storage density, are non-flammable, inexpensive, and have higher thermal conductivity than organic PCMs. However, they suffer from phase separation and supercooling [25,26]. Organic PCMs such as paraffin wax (PW), alcohols, fatty acids, esters, and polymer materials have high latent heat capacity, little or no supercooling, and are non-corrosive, but they have low intrinsic thermal conductivity. In this regard, the incorporation of thermally conductive nanomaterials into the PCM matrix can mitigate this problem [27,28]. However, despite the advances in this research area, the low thermal conductivity of SLPCMs and the tendency of the liquid phase of SLPCMs to leak out of the system continue to hinder their usage in practical applications [29–31].

Being thermally conductive and compatible with organic PCMs, sp^2 -rich carbon-based nanomaterials are a class of filler material that can be added directly into PCMs to form phase change composites (PCCs) with improved overall thermal conductivity [32–35]. Increasing the thermal conductivity of PCMs is crucial as it helps to maintain a more uniform temperature distribution, thus minimizing the formation of hotspots and subsequently localized overheating. Furthermore, faster charging and discharging rates can be achieved, resulting in a more efficient storage and utilization of thermal energy. The sp^2 -rich allotropes of carbon nanomaterials consist mainly of graphene, carbon nanotubes (CNTs), expanded graphite (EG), and graphite nanoplatelets (GNPs). In recent years, the use of graphene and its derivatives such as graphene oxide (GO), reduced GO (rGO) nanosheets and aerogels has been widely investigated by researchers for the enhancement of PCM systems by leveraging their outstanding properties [36,37]. Compared to other PCC systems such as metal-organic framework (MOF)-based or nanoparticle-enhanced PCCs, graphene-based PCCs offer several advantages such as higher thermal conductivity, lightweight, mechanical reinforcement, and high surface area, making them suitable for applications where fast heat transfer and minimal weight are crucial. Due to the absence of or the low number of impurities that could affect the transport pathways for heat carriers, graphene and rGO possess higher thermal conductivities compared to GO. On the other hand, the oxygen functionalities present on the edges and basal plane of GO make GO easier to be dispersed in PCMs, while also being relatively easier to produce via the oxidation and exfoliation of graphite, compared to graphene and rGO. Finally, graphene-based aerogels and foams have very low densities and high specific surface areas. Having a porous and conductive network, these three-dimensional (3D) nanostructures not only offer an enhancement in thermal conductivity when mixed with PCMs but also provide the PCC with shape stability. Not only are shape-stable PCMs less prone to leakage during phase transition, but they can be encapsulated or molded into specific shapes, thus facilitating integration into existing systems such as building materials or thermal storage units. This ensures their long-term functionality and overall effectiveness across various PCM-based applications. As an alternative to being dispersed as a filler material, carbon-based nanomaterials can be used to encapsulate the PCM in a core-shell configuration [38,39]. In this way, the latent heat and phase transition temperature of the core-PCM can be regulated more efficiently with the enhanced thermal conductivity provided by the carbon-based shells. Moreover, the shell's

structural stability could endow the PCM with enhanced shape stability.

In this review, we summarize the recent developments in graphene-based PCCs for thermal energy storage (TES) applications with improved thermal efficiencies (Fig. 1). We discuss the different types of graphene-based filler materials that have been explored and their unique properties, ranging from pristine graphene to GO, rGO and functionalized graphene. We then look at how graphene-based filler materials can be incorporated into the PCCs as different types of structures – namely nanoparticles, porous nanostructures, and core-shell structures – and briefly talk about their inherent advantages and disadvantages. Lastly, we present our outlook on the current obstacles and possible future directions for graphene-based PCCs.

2. Nanoparticulate graphene-based fillers

2.1. Graphene

Graphene is a single layer of graphite and possesses a two-dimensional (2D) framework that closely resembles that of a hexagon or honeycomb structure. Since its discovery in 2004 [40], graphene has received huge attention from researchers and industries due to the unique sp^2 -hybridized structure of graphene, which gives rise to its high optical transparency, and superior structural, physicochemical, and electronic properties. As such, graphene and its derivatives have been used widely in fields like catalysis [41], optoelectronics [42], tissue engineering [43], and biosensors [44]. Graphene can be obtained through various ways, such as the mechanical exfoliation of graphite [45], the reduction of GO [46], chemical vapor deposition [47], and epitaxial growth [48]. Besides, it also has a thermal conductivity of around 3000–5000 W/(m·K), which is considerably higher than CNTs and copper [49,50]. Consequently, considering all its remarkable properties, graphene is a highly desirable filler material for PCMs, not only for the enhancement of thermal conductivity, but also contributing to the simultaneous improvement in the electrical conductivity and mechanical properties of the PCC.

Using the melting infiltration method, Zhu et al. fabricated a lauric acid (LA)/iron foam composite PCM (CPCM) by adding graphene nanoplatelets (GNPs) into the PCM matrix [51]. Different pore densities (40, 70, and 90 PPI) of iron foam were used. The SEM images of LA/iron foam and

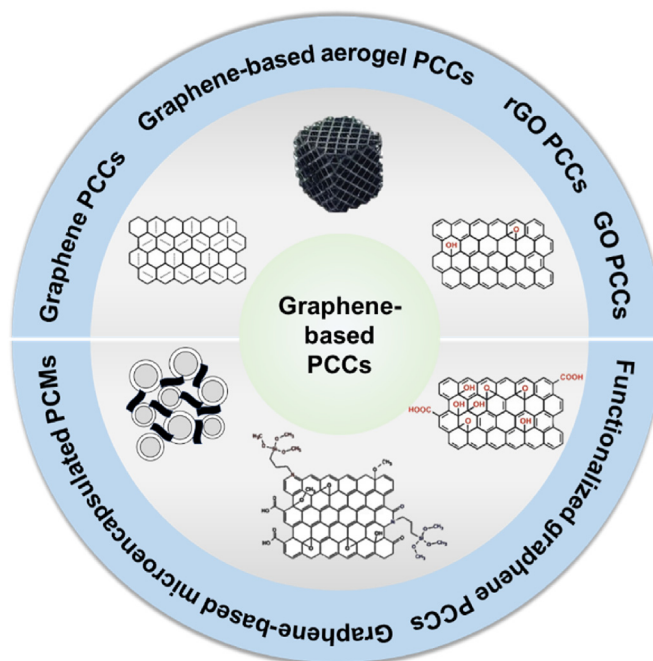


Fig. 1. Overview of graphene-based PCCs covered in this work.

LA/iron foam/GNP CPCMs of different pore densities are shown in Fig. 2a–f, showing that the GNPs were homogeneously dispersed within the iron foam matrix. Due to the layered structure of GNPs, the latter resulted in a smoother surface for the CPCM. It was found that the pore density of 70 PPI displayed the best impregnation effect among the three pore densities while there was an obvious enhancement in thermal conductivity with the addition of GNPs (Fig. 2g). Compared to non-porous LA/iron foam/GNP, the thermal conductivity of LA/iron foam/GNP CPCMs with 90 PPI pore density increased by 10.67 times, reaching a value of 1.227 W/(m·K). With the same pore density, the thermal conductivity of LA/iron foam/GNP CPCMs is higher than that of LA/iron foam without GNPs, confirming that the addition of GNPs had a significant effect in increasing the thermal conductivity. Furthermore, with a higher pore density of the iron foam, the average pore size was reduced. This led to a reduction in the interfacial area and a smaller air volume which increased the overall thermal conductivity.

Similarly, Chen et al. added GNPs to 1-hexadecanol (HD) and high-density polyethylene (HDPE) to improve the thermal conductivity of the obtained PCC [52]. Here, HD is used as the main PCM, whereas HDPE, as the matrix material, provides thermal stability, impact resistance, and prevents leakage of HD during the melting process as HD is converted from a solid to liquid phase. The PCC with 3 wt% GNP had a thermal conductivity of 0.67 W/(m·K), which was 1.86 times higher than that of HD/HDPE composite (0.36 W/(m·K)) without GNPs. Besides enhancing the thermal conductivity, GNPs also help to prevent the leakage of HD during the phase change process by strengthening the structure of the PCC. In another example, Prado and Lugo developed and synthesized a new nano-enhanced PCM based on dispersions of GNPs in a commercial stearate-based PCM, PT8 [53]. The stability of the dispersions was improved by adding acetic acid as a surfactant. An enhancement in the thermal conductivity of the nano-enhanced PCM by up to

52% was achieved compared to the base PCM, which also resulted in a higher thermal diffusivity (up to 32%). Meanwhile, Amin et al. investigated the thermal properties of a PCC where GNPs were added to a PCM of beeswax/graphene [54]. It was found that the addition of 0.3 wt% GNP drastically increased the thermal conductivity of the PCC from 0.25 to 2.89 W/(m·K), while the latent heat and heat capacity increased by 22.5% and 12%, respectively.

To enhance the properties of pure salts for TES applications, Xiao et al. prepared a nitrate- and nitrite-based molten salt mixture (denoted here as HITEC salt) that was doped with graphene flakes [55]. Subsequently, the molten salt/graphene mixture was impregnated into nickel and copper metal foams to form the PCC. The metal foams were found to be compatible with molten salt/graphene mixture based on the high impregnation ratio, while the thermal cycling results indicated good stability and reliability. With the presence of the graphene flakes and metal foams, the HITEC salt/graphene/metal foam PCCs had their thermal conductivities distinctly enhanced over the pure HITEC salt PCM, with an 1140% and 330% increment for the HITEC/3 wt% graphene/copper foam PCC and HITEC/3 wt% graphene/nickel foam PCC, respectively. It was noted that there was a significant decrease in latent heat of the salt/metal foam composites as compared to pure salt from 60.62 kJ/kg for the HITEC salt/nickel foam composite to 48.37 kJ/kg for pure HITEC salt. On the other hand, the addition of graphene only slightly affected the latent heat due to the small mass fraction of graphene and the high interfacial thermal resistance at the interface between the molten salt and graphene. The reduction of the specific heat of the PCC by metal foam could thus be compensated with the amount of graphene flakes present. In this regard, an optimization of the loading of graphene and metal foam is required to attain the desired thermophysical properties of the PCC.

Besides enhancing thermal conductivity, graphene can also be used to

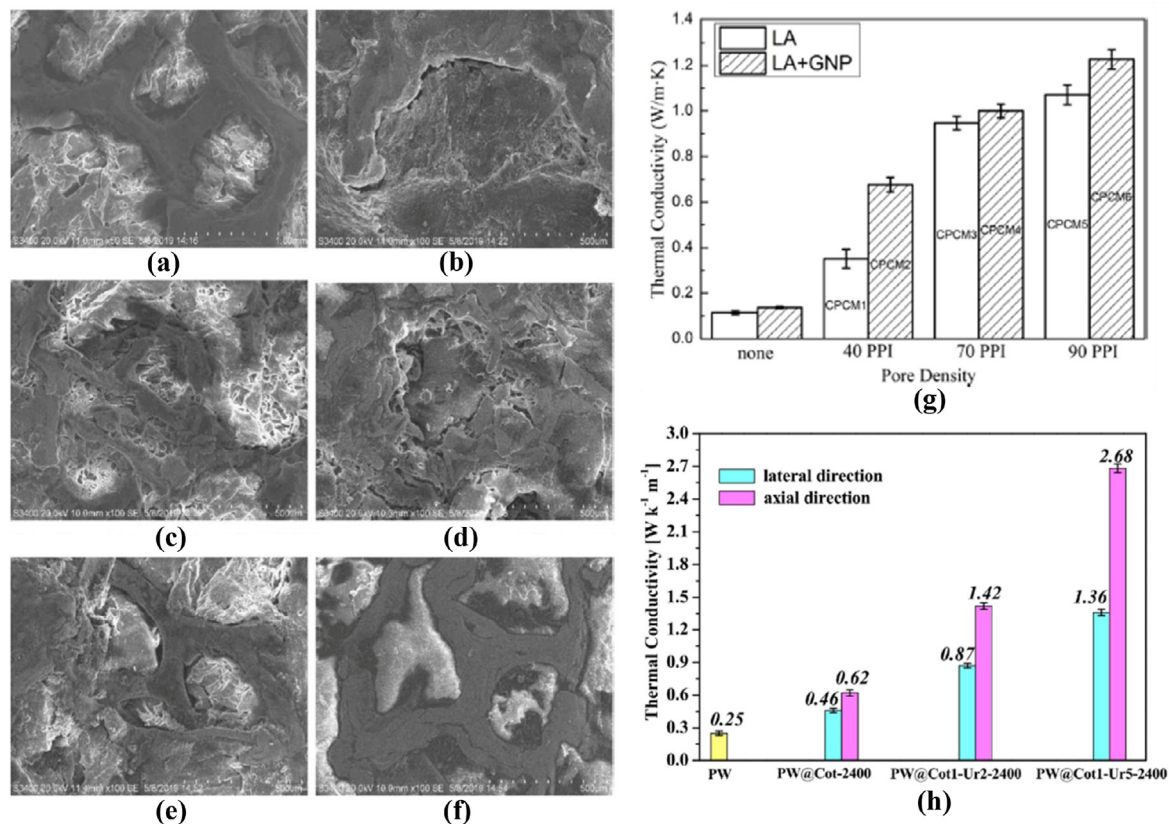


Fig. 2. SEM images of (a) LA, 40 PPI (CPCM1), (b) LA with GNPs, 40 PPI (CPCM2), (c) LA, 70 PPI (CPCM3), (d) LA with GNPs, 70 PPI (CPCM4), (e) LA, 90 PPI (CPCM5), and (f) LA with GNP, 90 PPI (CPCM6). (g) Thermal conductivities of the different CPCMs with different pore densities. Reproduced with permission [51]. Copyright 2020 Elsevier. (h) Thermal conductivities of the PCC samples. Reproduced with permission [56]. Copyright 2020 Elsevier.

improve the shape stability of SLPCMs, especially in the liquid phase. To solve the problems of leakage and poor thermal conductivity of organic PCMs such as PW, Sheng et al. worked on producing high-performance PCCs with high latent heat, high anisotropic thermal conductivity and good shape stability [56]. Vertically aligned hollow carbon fibers wrapped with graphene were used to construct biomass-derived porous carbon scaffolds which support the PCCs. Following that, a porous graphene framework (prepared by the facile *in situ* carbonization of urea on the cotton fiber surface) aided to enhance the thermal conductivity of the PW/porous carbon PCC. With a total carbon filler content of 8.5 wt%, the PCC exhibited anisotropic thermal conductivities of 1.36 W/(m·K) and 2.68 W/(m·K) along the lateral and axial fiber direction, respectively (Fig. 2h).

The incorporation of graphene as nanofillers into PCMs can not only enhance the thermal transfer but also the PCC's energy storage capability. However, graphene has a large specific surface area and therefore one should not exceed the optimum amount when adding this material as it might have adverse effects on the natural convection thermal conductivity of PCMs. In addition, the thickness and geometry of the graphene nanofiller play an important role in determining the overall thermal conductivity of the PCC and therefore is a key consideration when designing the desired PCCs.

2.2. Graphene oxide

GO, being an oxidized form of graphene, has a hexagon or honeycomb basal plane structure similar to graphene, but contains additional oxygen functionalities of hydroxyl, carboxyl, and epoxy groups on its edges and both sides of its basal plane. Due to hydrogen bonding between the oxygen functionalities and trapped water molecules in the basal plane, GO has a larger interlamellar spacing of 6–9 Å than graphene's 3.34 Å, and its interaction with other molecules can also improve its overall tribological and mechanical properties [57]. GO can be easily prepared from graphite by chemical and electrochemical methods, of which the Hummers and modified Hummers methods are most commonly used [58]. The outstanding properties of GO have led to its application in a broad range

of areas like catalysis [59], energy storage [60,61], and sensors [62].

Huang et al. prepared a shape-stabilized PCM (ssPCM) by adding GO to molten disodium hydrogen phosphate dodecahydrate (DHPD), an inorganic PCM. Thereafter, the DHPD-GO mixture was impregnated into expanded vermiculite (EV), a porous mineral that served to provide shape stability to the PCM [63]. By adding a low-weight fraction (0.2 wt %) of GO, the maximum latent heat of the composite increased from 167 J/g to 229 J/g. The presence of GO decreased the contact angle between melted DHPD and EV from 56 to 45°, indicating that GO improved the encapsulation volume of DHPD through the reduction of the interfacial tension between DHPD and EV. Further, as the carboxyl and hydroxyl groups on GO readily form hydrogen bonding with crystal water, this helped to reduce crystal water loss. The polar nature of the GO fillers was therefore beneficial in these two crucial aspects which contributed to the improvement in the thermal energy storage performance of the ssPCM.

Instead of impregnating the PCM into GO, Cao et al. covalently grafted hexadecyl acrylate (HDA) to GO via free radical polymerization to fabricate a ssPCM (GO-g-PHDA) [64]. Microscopic and spectroscopic studies confirmed the successful grafting of PHDA onto the surface of GO. The obtained ssPCM of GO-g-PHDA had a melting temperature and enthalpy of 35.0 °C and 79 J/g, respectively, at a GO loading of 17%. More importantly, GO-g-PHDA displayed excellent shape-stability compared to PHDA, with clearly no leakage or size change observed up to 80 °C (Fig. 3a).

Polymeric materials are a diverse class of substances that have attractive properties such as versatility [65], durability [66], and electrochromic [67] and are used in many applications [68–73]. Poly(ethylene glycol) (PEG) is a commonly used polymeric PCM as it offers a variety of molecular weights, high energy storage density, and can be applied in different fields [74–78]. Polymeric SSPCMs were fabricated by Xia et al., whereby GO served as the skeleton material, PEG was the PCM, and 4,4'-diphenylmethane diisocyanate (MDI) acted as the cross-linking agent [79]. Through graft co-polymerization and self-assembly approaches, the SSPCM displayed a regular, lamellar structure with PEG intercalated homogeneously into the GO nanosheets as determined by transmission electron microscopy (TEM) (Fig. 3b and c). With a loading

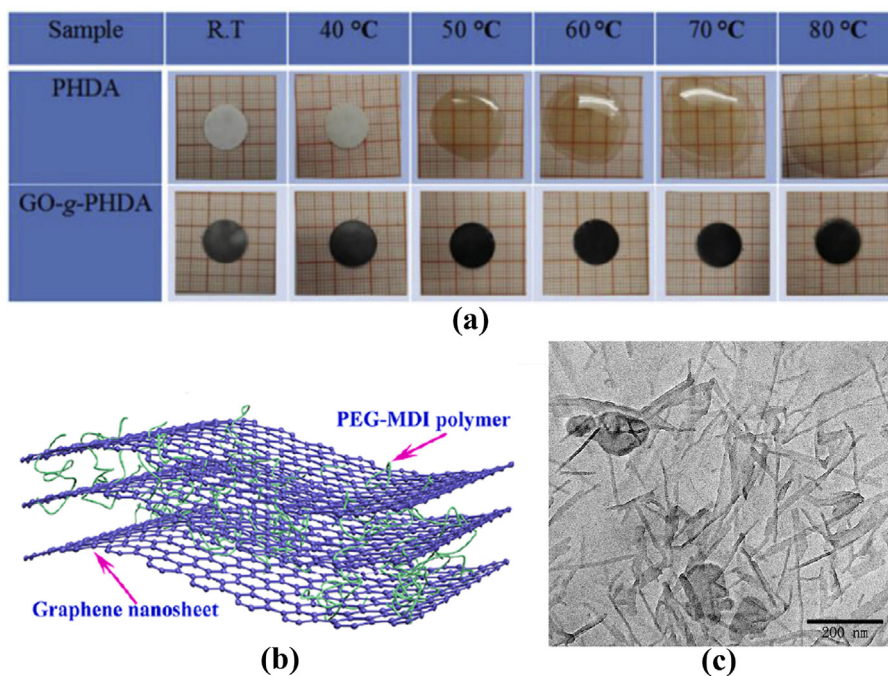


Fig. 3. (a) Photographs showing the shape-stabilities of PHDA and GO-g-PHDA with respect to temperature based on visual observation. Reproduced with permission [64]. Copyright 2017 Elsevier. (b) Schematic illustration for the graft-copolymerization of PEG-MDI and self-assembly process of the graphene nanosheets within the polymer. (c) TEM images of the SSPCM. Reproduced with permission [79]. Copyright 2019 Elsevier.

of 9 wt% GO, the thermal conductivity of the SSPCM increased by 41.4% from 0.338 to 0.478 W/(m·K) while the latent heat only decreased by 6% compared to pure PEG. Moreover, the ssPCM demonstrated good thermal cycling stability and could sustain a temperature of 50–57 °C for about 410 s in thermoregulation experiments.

Using a similar approach of confining PCMs within nanomaterials, Li et al. prepared a ssPCM by intercalating stearic acid (SA) into the inter-layer spaces of multilayer GO, enabled by interfacial interaction and capillary action [80]. Different SA/GO mass ratios were used and the authors found that the ssPCM with a 1:1 mass ratio prevented the leakage of SA. The thermal stability of the ssPCM was enhanced with the addition of GO as it served as a physical protective barrier on the surface. In terms of its thermal properties, the ssPCM maintained a relatively high thermal storage capability of 82.4% although its latent heat of 55.9 J/g was significantly reduced compared to pure SA's latent heat of 143.5 J/g.

2.3. Reduced graphene oxide

rGO is a form of GO that has undergone a reduction process to remove some or most of the oxygen-containing functional groups from its surface, resulting in a partially reduced material. This reduction process can be achieved through various methods such as chemical reduction using reducing agents like hydrazine [81] or hydriodic acid [82] or thermal treatment at high temperatures [83]. The reduction of GO leads to the partial restoration of the sp^2 hybridization of carbon atoms and the recovery of some of the electrical and mechanical properties of graphene.

Like graphene and GO, rGO has also been extensively explored as a thermally conductive and structurally supporting filler material in PCCs. Zhou et al. experimented with using a chromium-based MOF decorated with rGO as an adsorption material and structural support for a ssPCM

[84]. The MOF was first crosslinked with rGO using polyvinyl pyrrolidone and thereafter PW was introduced into the MOF composite via vacuum impregnation. A 60 wt% loading of PW in the composite gave the best form stability, with a latent heat and phase change temperature of 79.9 J/g and 51.5 °C, respectively. Compared to pure PW, the composite demonstrated an enhancement of 472% in thermal conductivity. Similarly, Wu et al. used the vacuum impregnation method with PW as the PCM to prepare a novel PCC with high latent heat, good PCM encapsulation ability, and thermal/light-actuated shape memory property [85]. As shown in Fig. 4a, GO sheets were first applied onto melamine foam (MF) by dip-coating, which aids in shape recovery and prevents PCM leakage. The GO sheets were then reduced by hydriodic acid to form a MF/rGO composite before being vacuum impregnated by PW. A high PW loading of approximately 2000 wt% was reported for the MF/rGO/PW PCC due to the low density and high porosity of the MF/rGO composite. The obtained PCC had a melting enthalpy of 144.8 J/g and exhibited excellent thermal reliability. In this work, rGO served as a light absorption medium which enhanced the solar-to-thermal conversion and thermal transport property of the PCC. Hence, this imparted the PCC with good shape stability and shape recovery when triggered by either light or heat.

Like vacuum impregnation, melt infiltration is a common approach for fabricating ssPCMs to overcome the leakage issue of liquid PCMs, where the PCM is infused into a mesoporous matrix that provides structural support to the PCMs [86]. However, one of the drawbacks of liquid infusion is that the physical interaction between the matrix and PCM might be too weak, leading to desorption issues after multiple thermal cycles. Chemical modification techniques like grafting, where the PCM material is covalently attached to the matrix, is an alternative approach that offers better thermal reliability and chemical stability [87,

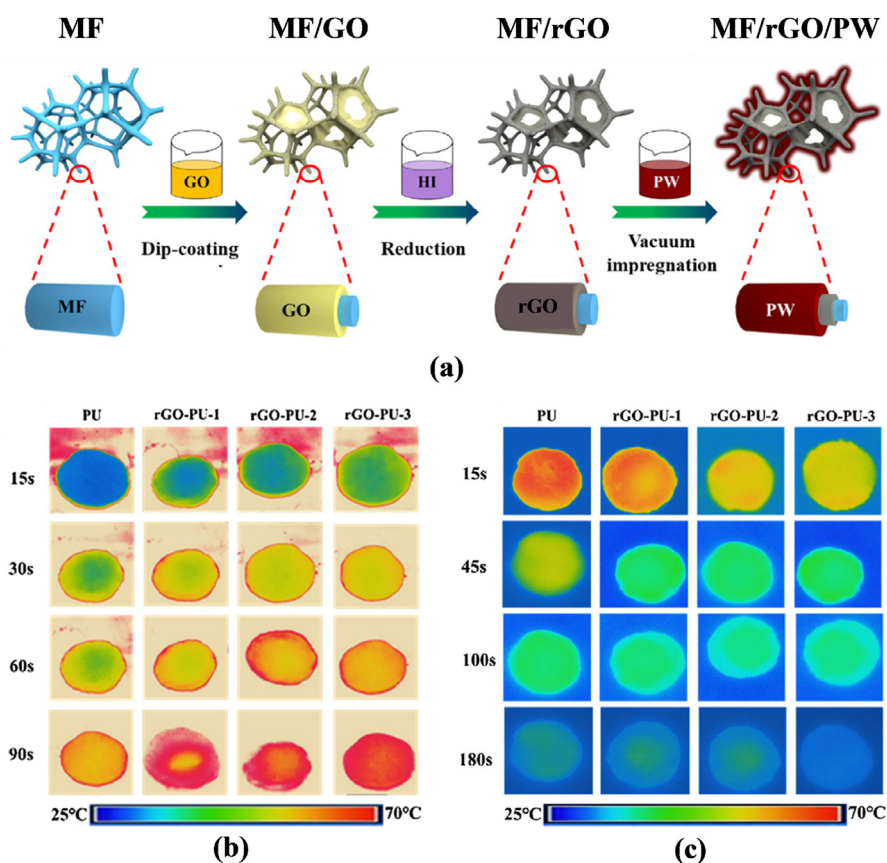


Fig. 4. (a) Schematic representations showing the preparation procedure of MF/rGO/PW PCM composites. Reproduced with permission [85]. Copyright 2020 Elsevier. Thermal map images of PU and rGO-PU SSPCMs showing the temperature distributions during the (b) heating process, and (c) cooling process. Reproduced with permission [89]. Copyright 2019 Elsevier.

[88]. Mu and Li fabricated a SSPCM where polyurethane (PU) was grafted onto rGO via PEG linkages (rGO-PU) by *in situ* polymerization, esterification, and reduction reactions [89]. Here the PEG linkages act as crystallizable soft segments that facilitate the phase change behavior or the SSPCM. The rGO-PU SSPCM possessed a melting enthalpy of 138.7 J/g and improved thermal stability, with a degradation temperature of approximately 24 °C and 4 °C higher than that of PU and PEG, respectively. The thermal conductivity of the rGO-PU SSPCM also reached 0.696 W/(m·K), which was 131.2% higher than PU. Infrared thermal images (Fig. 4b and c) showing the temperature distribution of the PU and rGO-PU SSPCMs revealed that the latter heated up and cooled down faster during the melting phase and cooling phase, respectively. The significant enhancement of the rGO-PU SSPCM's thermal conductivity was attributed to the rGO. In a similar study, Zhou et al. prepared PU-based SSPCMs by crosslinking GO nanosheets with PEG using hexamethylene diisocyanate (HDIB) via a facile solvothermal treatment [90]. During the crosslinking process, the GO nanosheets were reduced

simultaneously to rGO while ensuring excellent dispersity of the nanosheets in the PCM. A loading of 3.8 wt% GO in the SSPCM increased the phase change enthalpy from 86.8 J/g in pure PU to 127.3 J/g, and was sufficient to attain good thermal reproducibility and stability as well.

A summary of the works discussed in this section on nanoparticulate graphene-based fillers for PCCs is provided in Table 1. As we have seen, high-performance PCCs can be prepared via direct incorporation of graphene, GO, and rGO. Compared to other carbon-based materials such as CNT and EG, graphene has a better porous structure due to its two-dimensional structure with a high surface area and uniform pore size, possesses better chemical and thermal stability, and can provide shape stability to PCMs as well. The performance of graphene-based nanoparticulate fillers is contrasted to *n*-alkanes where the thermal conductivity of alkane-doped PCMs is not sufficiently high enough for practical applications despite having a long carbon chain. The addition of graphene nanofillers can dramatically increase the effective thermal conductivity of PCMs, but because of the tendency of graphene nanosheets to

Table 1
Summary of nanoparticulate graphene-based fillers used in PCCs.

PCC	Nanofiller loading (wt%)	Melting Temperature (°C)	Melting Enthalpy (J/g)	Solidifying Temperature (°C)	Solidifying enthalpy (J/g)	Thermal Conductivity (W/m·K)	Ref.
LA	–	45.5	174.1	41.6	179.4	0.12	[51]
LA/GNP	3	45.5	166.8	42.5	168.2	0.14	[51]
LA/GNP/iron foam (40 PPI)	3	45.7	170.5	42.1	172.8	0.68	[51]
LA/GNP/iron foam (70 PPI)	3	46.1	119.5	41.9	120.3	1.00	[51]
LA/GNP/iron foam (90 PPI)	3	46.1	94.4	42.1	95.2	1.23	[51]
1-HD	–	51.7	240.4	46.9	137.4	–	[52]
1-HD/GNP/HDPE	1	51.2	149.5	47.0	84.5	0.43	[52]
1-HD/GNP/HDPE	2	50.4	147.5	47.3	88.6	0.52	[52]
1-HD/GNP/HDPE	3	50.7	146.6	47.0	86.6	0.67	[52]
Beeswax	–	62.3	141.5	52.0	145.6	–	[54]
Beeswax/graphene	0.05	62.1	152.7	52.5	161.3	–	[54]
Beeswax/graphene	0.1	62.6	154.3	52.5	164.1	–	[54]
Beeswax/graphene	0.15	62.6	163.2	52.3	167.2	–	[54]
Beeswax/graphene	0.2	61.6	163.4	52.9	167.6	–	[54]
Beeswax/graphene	0.25	62.1	180.1	52.5	184.1	–	[54]
Beeswax/graphene	0.3	62.4	186.7	52.1	190.8	2.89	[54]
DHPD	–	43.6	240.0	–	–	–	[63]
DHPD/GO/EV	0.1	42.8	189.9	–	–	–	[63]
DHPD/GO/EV	0.2	42.1	191.4	–	–	–	[63]
DHPD/GO/EV	0.3	42.5	229.0	–	–	–	[63]
DHPD/GO/EV	0.4	42.2	204.0	–	–	–	[63]
GO-g-PHDA	17	35.0	79.0	30.5	77.0	–	[64]
PEG-8000	–	67.5	170.7	41.8	167.1	0.34	[79]
PEG-8000/GO	3	66.9	146.1	37.0	146.1	–	[79]
PEG-8000/GO	6	66.5	147.3	37.6	147.0	–	[79]
PEG-8000/GO	9	65.3	158.2	42.6	157.5	0.48	[79]
PEG-8000/GO	12	67.1	136.0	39.8	134.3	–	[79]
SA	–	74.0	140.9	57.0	143.5	–	[80]
SA/GO	–	33.0	55.7	23.0	55.9	–	[80]
SA/GO	–	36.0	56.9	21.0	59.0	–	[80]
SA/GO	–	32.0	56.1	23.0	53.4	–	[80]
PW	–	52.8	172.3	47.7	170.5	0.25	[84]
PW/rGO/MOF	5	51.5	79.9	47.4	77.4	1.43	[84]
PW	–	55.7	165.0	47.3	165.7	–	[85]
PW/MF	–	55.8	141.6	44.6	149.3	~0.062	[85]
PW/rGO/MF	5	56.8	136.3	44.2	150.9	0.096	[85]
PW/rGO/MF	10	57.8	144.8	44.1	154.1	–	[85]
PEG	–	44.6	180.5	35.2	176.1	–	[89]
PEG/rGO/PU	1	43.9	138.7	35.9	130.9	0.54	[89]
PEG/rGO/PU	3	42.7	132.9	36.1	127.6	0.61	[89]
PEG/rGO/PU	5	44.9	120.8	34.6	113.0	0.70	[89]
PEG	–	56.9	182.7	37.1	176.7	–	[90]
PEG/rGO/HDIB	0	48.4	86.8	24.2	85.8	–	[90]
PEG/rGO/HDIB	0.5	56.2	118.0	35.4	117.8	–	[90]
PEG/rGO/HDIB	1.0	57.5	123.2	38.1	121.1	–	[90]
PEG/rGO/HDIB	2.4	57.9	125.7	36.7	123.5	–	[90]
PEG/rGO/HDIB	3.8	56.5	127.3	37.3	125.5	–	[90]
PEG/rGO/HDIB	4.8	55.1	122.4	36.8	119.9	–	[90]
PEG/rGO/HDIB	9.1	55.9	107.2	37.4	103.6	–	[90]

agglomerate, it is challenging to obtain a homogeneous dispersion of graphene nanosheets when mixed with PCMs. This led to a huge interest in GO and rGO nanosheets, where they could be more easily dispersed, processed in an aqueous medium, and where the functional groups of GO allow further functionalization (which will be discussed in Section 2.4). Nevertheless, GO's drawback is its lower thermal and electrical conductivity due to the reduced amount of sp^2 -carbon in the basal plane. In addition, depending on how graphene and its derivatives affect the crystallinity of PCMs, the energy storage density of graphene-based PCCs may increase or decrease.

2.4. Functionalized graphene and its derivatives

Attaching functional groups to graphene and its derivatives could impart additional desirable properties such as dispersibility, stability, reactivity, and solubility [91]. There are several ways to functionalize graphene-based materials, and these include covalent functionalization, non-covalent functionalization and hybrid functionalization, which can be done *ex situ* or *in situ* [92,93]. Covalent functionalization entails the covalent attachment of functional groups directly to the carbon-rich network of graphene and its derivatives, and this can be carried out by processes like reduction, oxidation, click, and diazonium chemistry. On the other hand, non-covalent functionalization involves the attachment of foreign molecules or nanoparticles onto the carbon-rich network through non-covalent interactions, such as hydrogen bonding, van der Waals forces, π - π interactions, hydrophobic, and electrostatic interactions. Hybrid functionalization involves a combination of both covalent and non-covalent functionalization especially in cases where more than one functionality is added to the graphene-based material. These chemical functionalizations can be carried out during the synthesis of the

graphene-based material (*in situ*) or as a separate step after the graphene-based material has been obtained (*ex situ*). The following research examples illustrate some desirable properties that were attained by PCCs doped with functionalized graphene-based nanomaterial fillers.

Prabakaran et al. investigated a PCC of functionalized GNPs incorporated into a fatty acid-based PCM for air-conditioning application by studying its melting behavior [94]. To prevent the agglomeration of the graphene layers, covalent functionalization of GNPs with oxygen-rich functional groups such as $-\text{COOH}$, $-\text{OH}$ and $-\text{C}=\text{O}$ was carried out in a strongly acidic medium to produce a stable dispersion and enhanced stability of the nanocomposites. 0.1 to 0.5 vol% of the functionalized GNPs were prepared and it was found that with an increasing volume fraction of functionalized GNPs, the viscosity of the PCC increased as well. In addition, 0.5 vol% of GNPs was found to enhance the thermal conductivity of the PCC by $\sim 102\%$. A large average zeta potential value of $+43.248$ mV also indicated the long-term stability of the PCC. Depending on the operating load and the number of hours the air-conditioning unit had been operating, the source temperature from the air-conditioning unit could vary and affect the solidification of the PCM. This made the initial PCM temperature measured during the melting stage inconsistent. Thus, it was necessary to study the effect of the initial PCM temperature, T_{int} , on the melting process. When T_{int} was reduced from 2 to -10 °C, the melting process took a 23.9% longer time without the functionalized GNPs. With functionalized GNP loadings of 0.1, 0.2, 0.3, 0.4, and 0.5 vol%, the difference in the melting process time was reduced to 22.1%, 20.3%, 19.2%, and 17.6%, respectively. Essentially, as the volume percentage of thermally conductive functionalized GNPs in the PCC increased, the improved overall thermal conductivity of the PCC shortened the delay in the melting process. At T_{int} of -10 °C and 2 °C, the melting times of the PCC with 0.5 vol% of GNP were reduced by

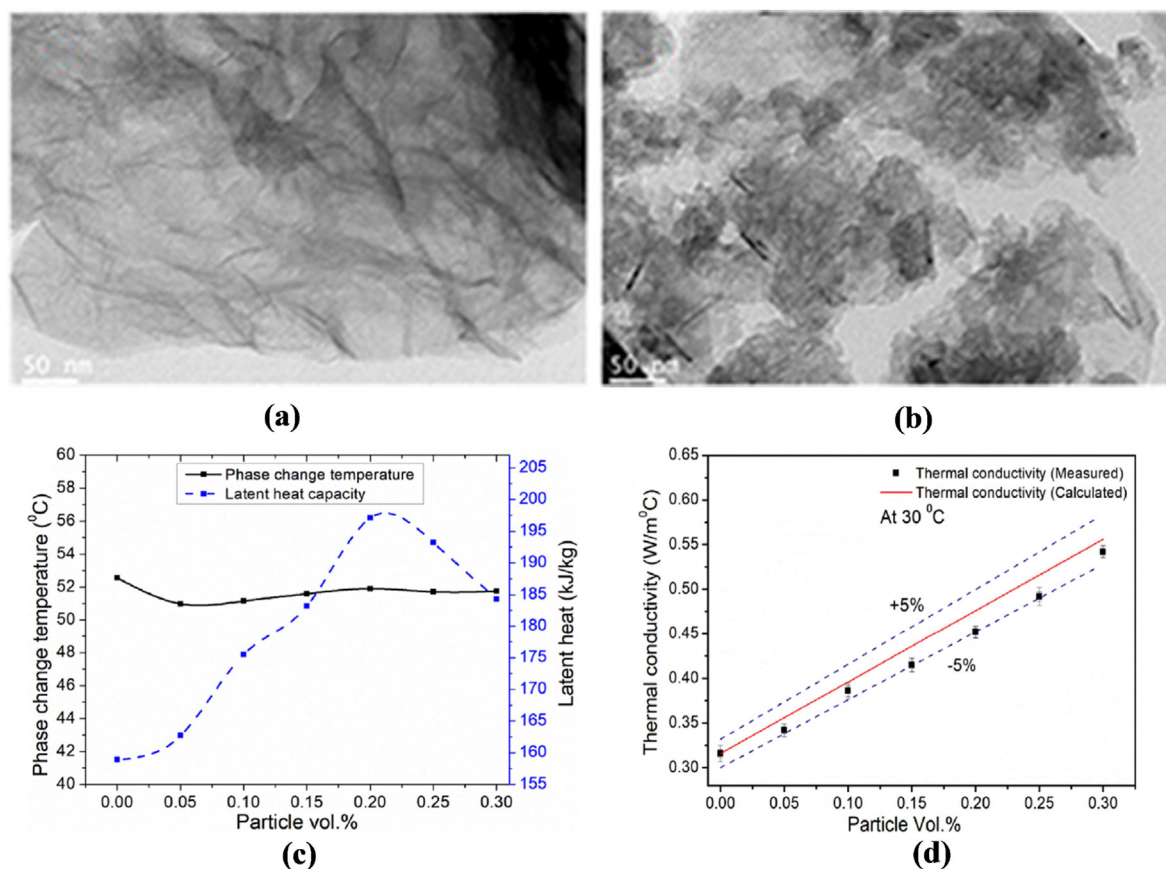


Fig. 5. TEM images of (a) pure graphene nanoplatelets, and (b) acidic functionalized graphene (AFG) nanoplatelets before being introduced into PEG to form a PCC. (c) Effect of nanoparticle concentration on the phase change temperature and latent heat capacity of the PCC, and (d) Effect of nanoparticle concentration on the thermal conductivity of the PCC at 30 °C. Reproduced with permission [95]. Copyright 2022 Elsevier.

up to 28.1% and 31.2% respectively, compared to the pure fatty acid-based PCM.

Selvaraj and Krishnan dispersed acidic functionalized graphene (AFG) in PEG to evaluate the PCC's thermal performance in a simulated electronic cooling system [95]. TEM images (Fig. 5a and b) revealed that the pure GnP had a transparent, thin flake-like structure while the acidic functionalization process, which introduced oxygen-rich functionalities to the GnP, did not affect the thickness of the AFG significantly. The PCC had a melting temperature range of 50–54 °C and melting enthalpy (latent heat capacity) of 158–200 kJ/kg. Compared to pure PEG, the PCC exhibited an enhancement in the latent heat capacity by 24% for a loading of 0.2 vol% due to the increased interaction between the PEG molecules and the AFG nanoparticles (Fig. 5c). However, the latent heat capacity started to decrease as the AFG nanoparticle loading increased further as the contribution from the low latent heat capacity of the AFG nanoparticles themselves became more dominant. Nevertheless, the AFG nanoparticles facilitated the enhancement of the overall thermal conductivity of PEG, with a maximum enhancement of 72% at AFG loading of 0.3 vol% (Fig. 5d).

Tao et al. demonstrated the potential of using *n*-alkyl acrylates as PCMs for TES applications [96]. Using a less conventional way to prepare ssPCMs, the authors first carried out free-radical copolymerization of *n*-HDA with 4-acryloyloxybenzophenone (ABP) to form a copolymer containing UV-sensitive ketone groups. Separately, chemical functionalization of graphene with HDA was carried out to obtain HDA-modified graphene (GN16). Eventually, GN16 was added to the HDA-ABP copolymer, and upon UV curing of the mixture, the ssPCMs (SPHDBG) were obtained. The addition of GN16 into the highly crosslinked ssPCM enhanced the thermal properties. SPHDBG had a melting enthalpy of 76 J/g and a phase change temperature of 36 °C. Additionally, the ssPCM exhibited excellent thermal responsiveness, shape stability, and solar-to-thermal energy conversion performance.

Waste recycling and mitigating the environmental impact of waste plastics is a crucial component of sustainable development. Many efforts [97–101] have been carried out by researchers to promote environmental stewardship and pollution reduction, including the use of biodegradable polymers such as polylactic acid and polyhydroxyalkanoates [102–104]. In an attempt to convert waste plastics into thermal storage materials, Chavan et al. carried out numerical and experimental analyses on linear low-density polyethylene (LLDPE), a common waste plastic material, by blending them with functionalized graphene (1, 3, and 5 wt%) and studied their thermal properties [105]. The thermal storage materials exhibited phase change behavior within a temperature range of 123–125 °C, and possessed heat of fusion values of 71.95–97 kJ/kg. In terms of their energy absorption capabilities during the melting process, LLDPE with 3 wt% functionalized graphene loading was able to store the most amount of thermal energy with an average of 1662 kJ/kg stored after 4000 s of melting, significantly higher than LLDPE without any functionalized graphene (1279.45 kJ/kg). This study thus showed that a waste plastic material such as LLDPE could potentially be re-used as thermal storage materials, thereby giving it a useful second life.

A novel photo-thermal PCC was fabricated by Li and Wang by grafting amino azobenzene (AAZO) derivatives onto GO and thereafter mixing them with PEG via ultrasound-assisted physical blending [106]. SEM images revealed that PEG was chemisorbed onto the surface of AAZO-GO by hydrogen bonding between PEG and the hydroxyl groups of AAZO and GO. The AAZO-GO/PEG PCC was found to have a high melting enthalpy of >86.5 J/g. Furthermore, due to the excellent thermal conductivity of AAZO-GO, the melting behavior of PEG was enhanced. Under visible light illumination, UV–vis spectra showed that the AAZO-GO/PEG PCC had an intense and broad absorption peak in the 380–470 nm range with an overall high absorbance in the visible range. Due to its good visible light absorbance, the AAZO-GO/PEG PCC demonstrated a high photo-thermal conversion efficiency of 91.0%.

By adding functionalized GO to PEG, Wang et al. prepared a SSPCM to

address the issues of low thermal conductivity, stability, and leakage [107]. The compound 2-amino-4,6-dichloride-1,3,5-triazine (ADCT) was first synthesized for the purpose of functionalizing GO to obtain the amine-terminated GO-based supporting material (GO-ADCT-NH₂). A copolymer of PU and PEG was then grafted on the functionalized GO by co-polymerization to obtain the PCC. With 9.6 wt% of GO-ADCT-NH₂, the latent heat only decreased by 5.3% while the thermal decomposition temperature increased by 38 °C compared to PEG. The CPCM also saw a 111% higher thermal conductivity than PEG. Additionally, GO-ADCT-NH₂ imparted the CPCM with flame retardant properties, reducing the heat release rate of the CPCM by 33.4% and delaying the burning time of the CPCM by 36 s according to microscale combustion calorimetry tests.

Meanwhile, Li et al. compared the thermal properties of polymeric PCCs comprising poly(ethylene-*graft*-maleic anhydride)-*g*-octadecanol (EMC18) as the PCM and either octadecylamine (ODA)-functionalized graphene oxide (GO-C18) or GO as the filler material [108]. GO-C18 and GO were incorporated into the PCC to function as shape stabilizers for the PCM. Expectedly, the shape stabilities of EMC18@GO-C18 and EMC18@GO were significantly enhanced due to the 2D framework of GO nanosheets. Good thermal reliabilities were also reported for both PCCs after 500 thermal cycles. However, due to the non-polar octadecyl side chains of GO-C18, the GO-C18 fillers presented better interfacial compatibility and interaction with the octadecyl chains in EMC18 than GO. As a result, EMC18@GO-C18 gave better shape stability than EMC18-GO, with no PCM leakage up to 160 °C.

Similarly, Ge et al. prepared CPCMs using epoxidized methoxy polyethylene glycol (EmPEG) as the PCM and polydopamine (PDA)-functionalized rGO as the filler material. The performance of this CPCM was compared with EmPEG incorporated with unfunctionalized (pristine) GO [109]. PDA-rGO and EmPEG were prepared and a one-pot reactive blending process was used to fabricate the EmPEG/PDA-rGO CPCMs (Fig. 6a). Compared to EmPEG and EmPEG/GO, EmPEG/PDA-rGO CPCMs were found to have higher thermal reliability, shape stabilization, and structure stability due to the more extensive interfacial interactions (polar interactions and hydrogen bonding) of the PDA-rGO nanosheets with EmPEG. A 3 wt% PDA-rGO endowed the CPCM with good stability after 100 thermal cycles and no PCM leakage was reported up to 105 °C.

To simplify the procedure for preparing carbonaceous CPCMs by eliminating steps like impregnation and freeze-drying, Akhiani et al. prepared a CPCM using palmitic acid (PA) as the PCM and oleylamine-functionalized rGO (OA-rGO) as the filler material, via a one-step self-assembly process [110]. The epoxy and carboxyl groups present in GO readily reacted with the amine group of OA, which reduced the inter-GO repulsion force and resulted in a hydrophobic OA-rGO surface. The OA-rGO facilitated the absorption and self-assembly of PA between the graphene nanosheets gel network owing to the hydrophobic van der Waals interactions between the PA hydrocarbon chains and the OA hydrocarbon chains of OA-rGO (Fig. 6b). A high phase transition enthalpy of 196.6 J/g and good shape stability was reported for the CPCM with a filler loading of only 0.6 wt% OA-rGO. Whereas, with 2 wt% loading of OA-rGO, the thermal conductivity was enhanced by 1.5 times.

Table 2 summarizes the thermal properties of PCCs containing functionalized graphene-based materials as fillers discussed herein. To select the most suitable functionalization method, the desired properties and intended applications of the PCCs need to be considered as each method has its advantages and limitations in terms of efficiency, scalability, and reproducibility. For example, covalent functionalization provides a strong attachment of the functional groups to the graphene backbone and offers precise control over the filler loading densities. On the other hand, non-covalent functionalization may result in weak, reversible binding of functional molecules and low filler densities, but it is relatively versatile and simple to carry out. Furthermore, the intrinsic properties of graphene will not be affected by the introduction of defects, which is unavoidable in covalent functionalization. Hybrid functionalization provides a

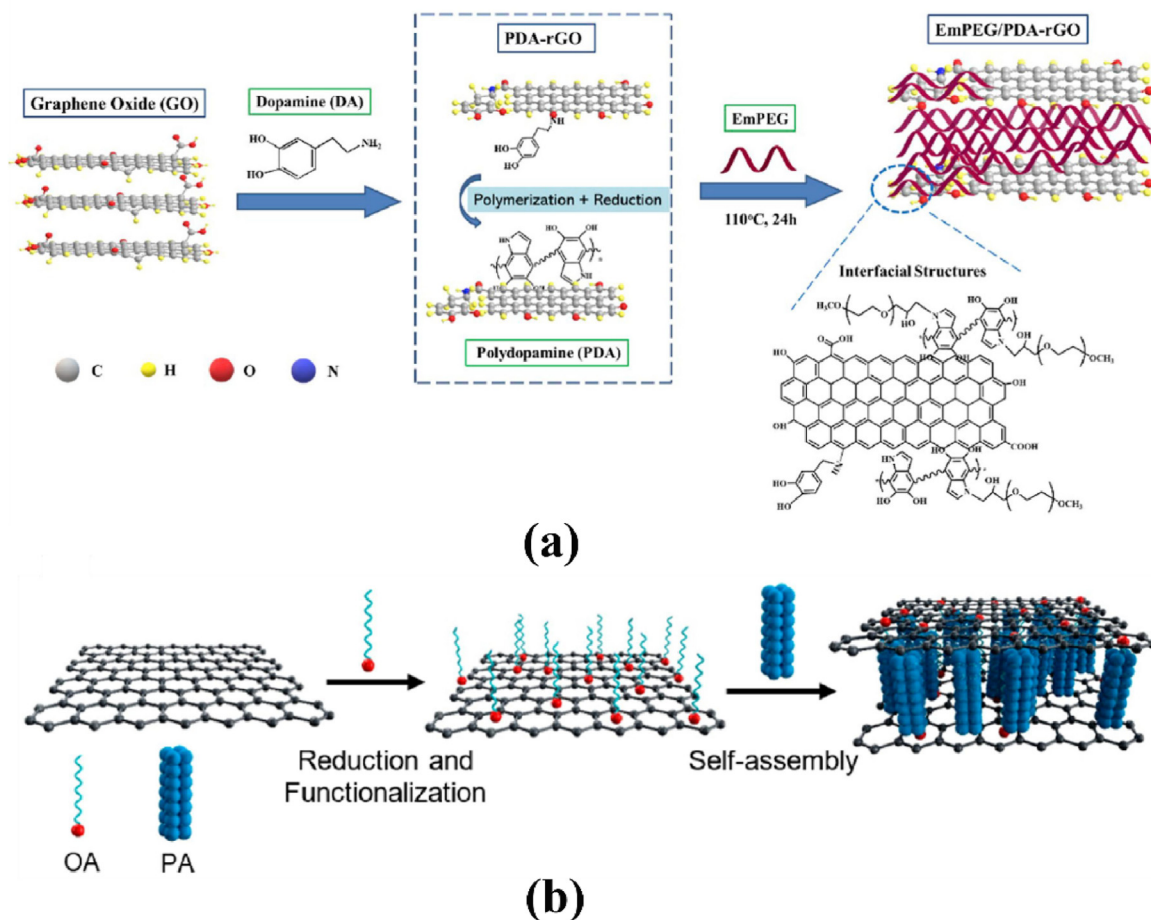


Fig. 6. (a) One-pot preparation process of EmPEG/PDA-rGO CPCMs. Reproduced with permission [109]. Copyright 2020 Elsevier. (b) Reduction, functionalization, and self-assembly process for preparation of the PA/OA-rGO CPCM. Reproduced with permission [110]. Copyright 2015 American Chemical Society.

combination of the benefits of both covalent and non-covalent functionalization but multiple steps and processing are required and defects and impurities may be introduced. Overall, the trade-offs of each method must be considered to achieve the desired properties and performance.

3. Porous nanostructured graphene-based fillers

As mentioned in the introduction, carbonaceous nanomaterials are added to PCMs for thermal conductivity enhancement. Porous nanostructures, in particular, are especially advantageous because, as scaffolds, they can additionally significantly improve the shape stability of the PCC [111,112]. Porous nanostructures can be classified into carbon aerogels and graphene aerogels (GAs). Carbon aerogels have high surface areas, are lightweight, and possess relatively high electrical conductivity. They are therefore ideal for use as electrodes in supercapacitors and batteries, catalysis and insulation applications [113–115]. GAs, being a type of carbon aerogel, also have high thermal conductivities and excellent mechanical strength, and apart from their use in PCCs, they have also been explored for many other applications [116–118]. By providing a scaffold for PCMs to maintain their shape during the phase transition, GAs can be used to prepare form-stable PCMs (FSPCMs), and their high porosity and surface area can maximize the heat transfer to and from the PCM, making the PCC more efficient.

Cao and Zhang prepared GA-based PCCs using the hydrothermal method and investigated the effects of GA in the composite [119]. As the content of GA increased, the crystallinity of PW increased from 46.21 to 61.11% due to the nucleation sites provided by the oxygen-containing functional groups present in GA which promote crystallization. With

9.2 wt% of GA added, the latent heat increased from 207.9 to 222.5 kJ/kg and the thermal conductivity was able to reach 0.81 W/(m·K), an increment of 286% compared to PW. This is due to the high thermal conductivity and continuous network of GA. The high thermal conductivity of GA was found to have a positive influence on the activation energy needed for PW's phase change. A loading of more than 7.3 wt% GA greatly reduced the activation energy, resulting in an improved thermal response rate of the composite.

A one-step fixed-point heat flow method was used by Huang et al. to prepare rGO/boron nitride (BN) anisotropic aerogels (CGAs) to form PCCs by vacuum adsorption of PW [120]. In addition to rGO, BN was introduced to further increase the thermal conductivity and improve the rigidity of CGA to reduce its contraction during the cooling cycle. An axial thermal conductivity of 1.68 W/(m·K) was obtained when a mass ratio of 1:20 of GO/BN was used to prepare the CGA, which is 504% higher than a pure PW PCM. With the CGA, the rate of PW leakage and the enthalpy loss rate was low at 3.1% and 2.7% respectively after 50 cycles. Furthermore, despite repeated melting and solidification of PW, the integrity of the heat conduction path was maintained. Impressively, due to the efficient thermal conduction of the CGA network, the PW phase change time was shortened by up to 1210 s, giving an enhancement in the phase change efficiency by approximately 47% when simulating an actual temperature change. As compared to pure PW, the phase transition onset of PCC occurred earlier, and therefore less time was required to complete the phase transition. Overall, the CGA greatly helped in improving the thermal conduction, thermal response and hence the efficiency of the phase change energy storage in PW.

Zheng et al. prepared a novel FSPCM by encapsulating paraffin in the

Table 2

Summary of the thermal properties of PCCs containing nanoparticulate functionalized graphene-based fillers.

PCC	Preparation method	Nanofiller loading (wt%*/vol%**)	Melting Temperature (°C)	Melting Enthalpy (J/g)	Solidifying Temperature (°C)	Solidifying enthalpy (J/g)	Thermal Conductivity (W/m·K)	Ref.
OM 08	• Direct mixing	–	8–10	180.0	8–10	–	0.24	[94]
OM 08/GnP		0.5**	–	–	–	–	0.44	[94]
PEG-4000	• Melt-blending	–	53–58	158.9	–	–	0.32	[95]
PEG-4000/graphene		0.2**	–	197.1	–	–	–	[95]
PEG-4000/graphene		0.3**	–	184.3	–	–	0.54	[95]
PHDA	• Direct mixing, UV curing	–	42.0	86.0	24.6	83.0	–	[96]
PHDA/graphene		5*	36.0	76.0	25.8	75.0	–	[96]
LLDPE	• Melt-blending	–	123.0	71.8	–	–	0.32	[105]
LLDPE/graphene		1*	122.5	91.8	–	–	0.39	[105]
LLDPE/graphene		3*	122.0	92.9	–	–	0.88	[105]
LLDPE/graphene		5*	120.0	97.0	–	–	0.89	[105]
PEG	• Direct mixing	–	51.1	135.8	32.7	122.8	–	[106]
PEG/GO		5*	50.5	89.4	31.7	77.8	–	[106]
PEG/GO		10*	50.6	87.6	31.6	76.7	–	[106]
PEG/GO		15*	50.5	86.5	31.1	75.3	–	[106]
PEG-6000	• Direct mixing	–	61.4	184.2	43.0	174.4	0.47	[107]
PEG-6000/GO		9.6*	60.2	157.6	38.4	141.8	0.98	[107]
EMC18	• Direct mixing	–	58.7	150.9	43.7–53.5	146.8	–	[108]
EMC18/GO		0.5*	58.6	150.1	42.6–53.4	146.0	–	[108]
EMC18/GO		1*	58.8	149.3	42.5–53.2	145.5	–	[108]
EMC18/GO		2*	59.1	147.8	42.5–53.0	143.9	–	[108]
EMC18/GO-C18		0.5*	49.8–59.1	150.3	40.5–53.2	146.2	–	[108]
EMC18/GO-C18		1*	50.0–59.3	149.8	40.5–53.2	145.7	–	[108]
EMC18/GO-C18		2*	50.2–59.8	148.6	39.2–52.8	144.6	–	[108]
EmPEG	• Direct mixing	–	60.0	179.6	40.2	179.6	0.39	[109]
EmPEG/PDA-rGO		0.5*	60.3	168.2	39.6	163.7	–	[109]
EmPEG/PDA-rGO		1*	60.4	165.0	41.5	163.2	0.48	[109]
EmPEG/PDA-rGO		2*	60.1	164.4	40.3	161.4	–	[109]
EmPEG/PDA-rGO		3*	60.4	160.6	42.0	158.3	0.52	[109]
EmPEG/PDA-rGO		4*	60.2	158.1	44.8	155.4	–	[109]
PA	• Direct mixing	–	64.3	203.0	58.9	214.1	~0.28	[110]
PA/OA-rGO		0.6*	64.1	196.6	59.4	203.0	0.29	[110]
PA/OA-rGO		1*	64.3	187.5	59.3	194.3	0.31	[110]
PA/OA-rGO		1.5*	64.7	184.9	59.0	190.0	0.32	[110]
PA/OA-rGO		2*	64.8	181.9	59.1	189.3	0.42	[110]

matrix of a copper foam (CF) [121]. GA was further added to prevent leakage of the paraffin and to increase the overall thermal conductivity. SEM analysis revealed that GA was successfully incorporated into the CF without damaging the skeleton (Fig. 7a–f). Moreover, the micropores of GA were completely impregnated with paraffin. Leakage tests conducted after repeated thermal cycles revealed that the leakage rate increased with the number of thermal cycles, although a plateau in the leakage rate was reached after an equilibrium state of mass transfer had been achieved. Nevertheless, it was noted that increasing the concentration of GO from 10 mg/mL to 30 mg/mL significantly improved the leakage performance. Owing to the thermally conductive interconnected structure of CF as well as the high thermal conductivity of GA, the composites exhibited more than a nine-fold increase in thermal conductivity compared to paraffin.

Xue et al. fabricated a hybrid GnP aerogel with in-plane walls that are compact and stacked oriented in one direction, connected by many through-plane bridges perpendicular to these walls [122]. The hybrid

GnP aerogel was fabricated via co-mediated assembly of GnPs in a MF and cellulose nanofiber (CNF) network using an ice-templating method. The resultant 3D-GnP network was intended to enhance the PCM encapsulation capability and thermal conductivity. CNF aided in the dispersion of GnPs while the imino groups of MF facilitated the interaction of CNF and MF via hydrogen bonds. Compared with a similar PCC using only GA as a filler, the PCC with the hybrid GnP aerogel had better encapsulation, thermal cycling, and thermal stability. In addition, compared to pure PW, the thermal conductivity of the PCC with the hybrid GnP aerogel filler was enhanced by 407% with a GnP content of 4.1 wt%.

Even though GA is considered a high-volumetric material with a porous structure, its structural stability is a concern in the event of volume shrinkage. Recognizing this issue, Yu et al. fabricated a modified GA by crosslinking GA with cysteamine [123]. The graphene/cysteamine aerogel was prepared by permeating cysteamine vapor through the GA. The obtained aerogel had a high surface area, 3D porous structure, and

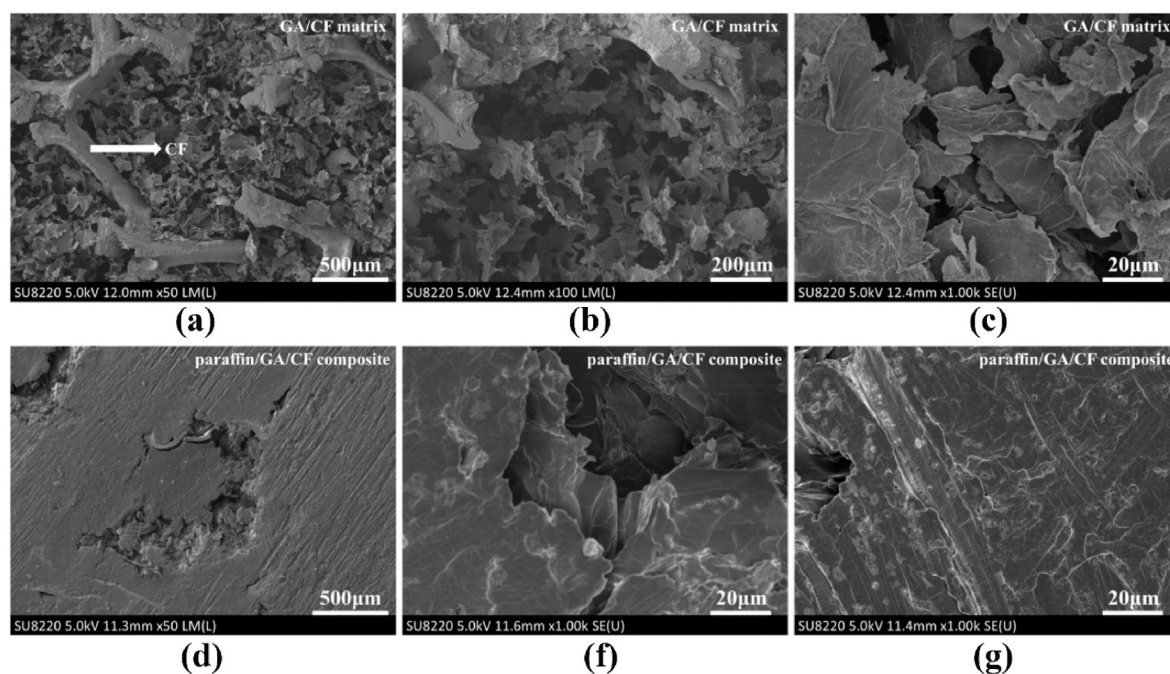


Fig. 7. SEM images of (a–c) the GA/CF matrix, and (d–f) the paraffin/GA/CF composite PCM. Reproduced with permission [121]. Copyright 2021 Elsevier.

could contain a large amount of PCM in the internal structure without volume shrinkage. Similarly, Wei et al. introduced lignin as a modifier into GA, which extended and supported the GA skeleton to reduce volume shrinkage [124]. Fig. 8a shows the images of GA and the lignin-modified GA (LGA) samples before freeze-drying. Compared to the LGA samples, the GA exhibited a large shrinkage effect during freeze-drying due to the aggregation of graphene nanosheets, while the GA samples with a higher content of lignin resulted in less shrinkage (Fig. 8b and c). From the SEM analysis, GA does not have a well-defined 3D structure, and the arrangement of the layer-by-layer graphene sheets is irregular (Fig. 8d). On the other hand, LGA with 1 wt% lignin (LGA1.0) has an interconnected and well-defined 3D network due to noncovalent interactions between lignin and GA (Fig. 8e). With the incorporation of PEG into LGA1.0, the crowded pores can contain the PCM by capillary

force. As a result, the composite has a smooth surface and an almost featureless structure (Fig. 8f and g).

GO aerogels are formed from the self-assembly of GO sheets to construct a 3D framework with high porosity and ultra-low density. In addition, as PCMs like PEG can form hydrogen bonds with the oxygen-containing functional groups on the GO aerogel, it is advantageous for the impregnation of PEG by capillary force into the GO aerogel pores, thus preventing leakage. Using this technique, Bao et al. prepared FSPCMs for solar thermal applications [125]. Due to the lower degree of graphitization of GO, the color appeared yellow instead of graphitic black. CNTs and carbon spheres (CSs) containing sp^2 -rich graphitic carbon were thus added to increase the blackness of the GO aerogel to enhance sunlight absorption. By doing so, a photothermal conversion efficiency of 89.3% was reached. The porosity, high surface area and high

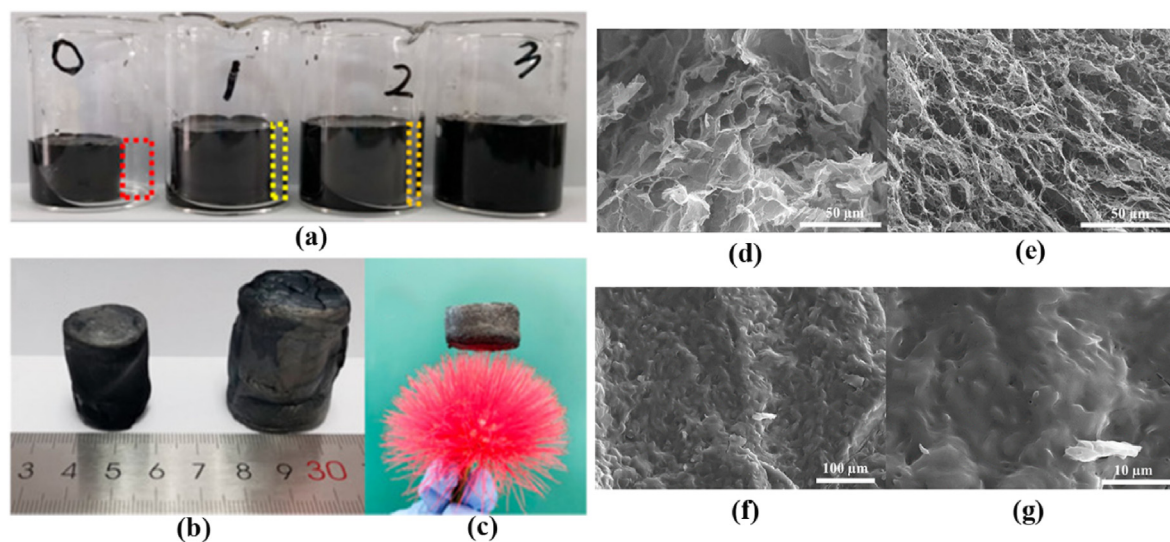


Fig. 8. Digital photographs of (a) pure GA, LGA0.5, LGA1.0, LGA1.5 before freeze – drying (left to right), (b) GA and LGA 1.0 (left to right), (c) LGA1.0 on red powder puff flower. SEM images of the fractured surfaces of (d) GA, (e) LGA1.0, (f) PEG@LGA1.0, and (g) PEG@LGA1.0 at higher magnification. Reproduced with permission [124]. Copyright 2021 Elsevier.

retention of PEG in the GO aerogel gave rise to a high PEG absorption capacity of 96.4% for the FSPCM. At the same time, the incorporation of the CNT and CS nano-additives contributed to boosting the thermal conductivity by 181.58% over pure PEG, while also displaying good thermal stability and good form stability after 12 h at 80 °C.

The structure and formation of GO aerogels from GO are influenced by the oxidation level of GO, but this correlation is rarely investigated. Therefore, Tang et al. introduced PEG into GO aerogels prepared from GO with varying degrees of oxidation and studied their structures and form stabilities [126]. It was concluded that with increasing oxidation levels, more oxygenated functional groups were present in the GO aerogel network. Moreover, at higher degrees of oxidation, more epoxy and carboxyl groups started to form over hydroxyl groups. Due to the disruption of the graphitic stacking order, the degree of graphitization of GO decreased while the sp^3 domains of GO increased. A higher oxidation level also resulted in better shape stability for the composite as there were more polar interactions between PEG and the oxygen-rich functionalities of the GO aerogel, leading to a higher capillary force.

Meanwhile, Xi et al. prepared a rGO aerogel using the hydrothermal-freeze drying method and added oxygen-deficient TiO_2 (TiO_{2-x}) to enhance the thermal conductivity and optical performance of PCCs [127]. A highly efficient heat transfer pathway provided by the rGO network in the composite material matrix enabled rapid and uniform heating of each part of the rGO@PW PCC material. This significantly improved the thermal conductivity by 128.6% compared to PW. The addition of TiO_{2-x} nanoparticles content of 100% further increased the thermal conductivity by 52.5% as well. The TiO_{2-x} /rGO@PW composite also possessed good light absorbance across the UV, visible and near-infrared ranges, which is beneficial for photothermal conversion. At an optimum TiO_{2-x} content of 80%, the PCC could attain 89.9% photo-thermal conversion efficiency.

As mentioned earlier, graphene-based aerogels can be used to provide shape stability and prevent leakage of PCMs. Nevertheless, the ability of these aerogels to prevent leakage depends on their pore structure, whereby their pore sizes can range from a few to hundreds of micrometers. Therefore, to restrain the leakage of PCMs from the pore structures of rGO aerogels, Cai et al. went a step further by adding a thickening agent of poly(styrene-*b*-ethylene-co-butylene-*b*-styrene) (SEBS) to adjust the viscosity of PW [128]. This is in accordance with several studies which have shown that SEBS could improve the shape stability of PCMs [129,130]. Due to the large pore size of rGO aerogel, the leakage test result was undesirable without SEBS, but when 5 wt% to 15 wt% of SEBS were used, no leakage of the PW was observed. Furthermore, a high latent heat of 226 J/g and a good thermal cycling stability (decrease in enthalpy by 3% after 200 thermal cycles) were achieved with 5 wt% SEBS.

A summary of the PCCs containing porous nanostructured graphene-based fillers, primarily graphene-based aerogels, is presented in Table 3. The use of graphene-based aerogels as matrices for PCMs generally enhances the thermal performance of the PCCs by providing an efficient and stable platform for energy storage and thermal transfer to and from the PCM. In addition, the graphene-based aerogel provides the PCC with a shape-stable and thermally conductive porous scaffold that can be used to effectively prevent PCM leakage. PCMs with graphene-based aerogel systems can be fabricated via different methods, such as hydrothermal, sol-gel, electrospinning, freeze-drying, and co-precipitation. The cost of materials and complexity of the process for each of these methods are different, as well as the resulting properties of the composite.

4. PCMs encapsulated with graphene-based materials

Encapsulation is typically used to confer to the system properties such as protection, stability and controlled release, and is therefore important for a wide range of applications [131–135]. The microencapsulation of PCMs in the form of a core-shell structure can not only provide shape stability but also prevent the leakage of these materials. However, the

issue of this method lies with the generally low thermal conductivity of the core-shell morphology [136,137], which would translate to a low thermal conductivity of the PCC material itself. To address this issue, graphene, GO, or rGO shells can be used to encapsulate the PCMs. Graphene-based core-shell structures have many advantages, for example, the highly chemical resistant and superior mechanical properties of graphene can protect the core material from degradation and enhance the overall mechanical properties of the core-shell structure. Having a large surface area also allows for more efficient interfacial contact with the core for better thermal dissipation. Lastly, the shell material and the shell thickness (e.g., using multilayers or composite shell materials) can be tailored to suit different TES applications as well. Many of such graphene-based shells are prepared from Pickering emulsions, where the graphene-based nanomaterials tend to self-assemble at the interface between the PCM (disperse phase) and the solvent (continuous phase). Being a straightforward, one-step process, this is regarded as an efficient technique for the encapsulation of PCMs with graphene-based materials [138,139].

Chi et al. prepared a FSPCM by absorbing tetradecanol (TD) into an EG matrix and subsequently coating the TD/EG composite with GO via electrostatic self-assembly [140]. As shown in Fig. 9a, after immersing EG in molten TD, the TD/EG composite was wetted with 3-aminopropyl triethoxysilane-ethanol solution to obtain amino-modified NH_2 -TD/EG. Being positively charged in an aqueous GO dispersion, it attracts the negatively charged GO nanosheets to the interface, thus leading to the formation of a shell of agglomerated GO nanosheets. TD/EG-GO composite has a melting temperature of 35.2 °C and freezing point of 34.9 °C, with melting and freezing latent heat values of 189.5 J/g and 187.9 J/g respectively. After 500 thermal cycles, the latent heat of the TD/EG-GO composite decreased by less than 1%, clearly showing that the leakage of TD was effectively prevented by the combination of the EG structure and the GO shell. Moreover, heat transfer within the PCM was sped up by the 3D EG and GO shell network owing to their high thermal conductivity, surpassing even the TD/EG composite without GO encapsulation.

Liang et al. prepared a novel PCC of PW/EG@GO inspired by the sponge gourd's structure [141]. EG was impregnated into PW to form the core, and with GO as the shell, the synergistic effect of the core-shell structure significantly enhanced the PCC's thermal conductivity (1.33 W/(m·K)) (Fig. 9b) and shape stability. The PCC had a melting enthalpy of 158.5 J/g and was also thermally stable with excellent cycling reliability. The PCC was also tested for its temperature regulation capability, and the results indicated that it had a faster thermal response compared to epoxy resin (EP) and PW/EP during the heating process from 25 to 35 °C (Fig. 9c). The high thermal conductivity and fast thermal response of the PW/EG@GO PCC are attractive qualities for its use in TES applications such as for temperature control of electronic devices.

Similarly, Huang et al. constructed a 3D graphene skeleton via *in situ* polymerization to encapsulate the SLPCM of PW, and in doing so enhance its thermal conductivity and provide shape stability [142]. With a 15.1 wt% loading of graphene, an 300% enhancement in thermal conductivity compared to PW was attained, while maintaining a similar level of latent heat performance. In addition, the 3D framework provided the PCC an enhanced compressive strength of up to 410.83 kPa, good thermal cycling stability and prevented any leakage of the liquid PW.

Instead of constructing the core-shell using only graphene-based materials, composite shell materials where a graphene-based material is one of the shell components could also be used for encapsulation. For instance, Ren and Hao synthesized a microcapsule shell wall made from a composite of graphene and melamine formaldehyde resin (CGMF) as a barrier for PCMs [143]. This prevented the PCM from reacting with asphalt, the material in which the microcapsules were dispersed. Here, *n*-tetradecane was used as the liquid PCM core and the microcapsules were incorporated into asphalt by high-speed shearing. The CGMF shell was prepared by *in situ* polymerization of the melamine formaldehyde resin after different amounts of graphene were added. It was noted that there were no chemical reactions between graphene, melamine

Table 3

Summary of the thermal properties of PCM/graphene-based aerogel PCCs.

PCC	Preparation method	GO (mg/mL)	Melting Temperature (°C)	Melting Enthalpy (J/g)	Solidifying Temperature (°C)	Solidifying enthalpy (J/g)	Thermal Conductivity (W/m·K)	Ref.
Paraffin	• Hydrothermal, vacuum immersion	–	–	207.9	–	–	–	[119]
Paraffin/GA		4	–	–	–	–	–	[119]
Paraffin/GA		8	–	–	–	–	–	[119]
Paraffin/GA		12	–	–	–	–	–	[119]
Paraffin/GA		16	–	222.5	–	–	0.81	[119]
Paraffin	• Direct mixing, freeze drying, vacuum absorption	–	50–52	–	–	–	0.28	[120]
Paraffin/rGO/BN		–	55.8	162.1	49.1	–	–	[120]
Paraffin/rGO/BN		–	55.7	150.7	49.0	–	–	[120]
Paraffin/rGO/BN		–	55.9	145.2	48.7	–	–	[120]
Paraffin/rGO/BN		–	55.6	141.6	48.8	–	–	[120]
Paraffin/rGO/BN		–	55.7	126.5	48.7	–	1.68	[120]
PA60	• Direct mixing, freeze drying, vacuum impregnation	–	60.0	204.1	–	203.7	~0.30	[121]
PA70		–	72.2	266.1	–	263.8	~0.30	[121]
PA80		–	80.0	221.3	–	221.0	~0.30	[121]
PA90		–	92.0	213.3	–	207.5	~0.30	[121]
PA60/GA/CF		30	–	133.6	–	133.4	~3.00	[121]
PA70/GA/CF		30	–	174.6	–	173.1	~3.00	[121]
PA80/GA/CF		30	–	144.0	–	143.9	~3.00	[121]
PA90/GA/CF		30	–	140.7	–	136.9	~3.00	[121]
PW	• Direct mixing, freeze drying, vacuum-assisted encapsulation	–	51.7	150.4	–	–	0.28	[122]
PW/GnP aerogel/MF/CNF		–	52.4	147.9	–	–	1.42	[122]
PEG-4000	• Physical blending, hydrothermal-freeze drying, vacuum impregnation	–	61.1	171.6	41.5	165.4	–	[124]
PEG-4000/GA		–	61.9	148.5	41.1	147.3	–	[124]
PEG-4000/LGA0.5		–	61.0	165.8	41.2	161.0	–	[124]
PEG-4000/LGA1.0		–	61.0	168.1	42.0	163.6	–	[124]
PEG-4000/LGA1.5		–	61.1	168.7	41.9	165.1	–	[124]
PEG-10000	• Vacuum impregnation	–	63.2	154.8	43.2	149.1	0.38	[125]
PEG-10000/GO aerogel		9	62.7	150.2	42.5	144.5	~0.69	[125]
PEG-10000/GO aerogel/CNT/CS		9	62.1	148.4	41.9	141.6	~1.07	[125]
PEG-10000	• Vacuum impregnation	–	64.2	182.9	40.8	174.6	–	[126]
PEG-10000/GO aerogel		20	68.1	199.6	40.3	191.0	–	[126]
PEG-10000/GO aerogel		40	66.8	173.7	40.2	167.0	–	[126]
PW	• Hydrothermal-freeze drying, vacuum impregnation	–	55.9	176.7	45.6	171.9	0.35	[127]
PW/rGO aerogel		5	56.8	110.9	47.6	104.3	0.80	[127]
PW/rGO aerogel/30% TiO ₂		5	55.8	125.5	49.8	127.8	~0.91	[127]
PW/rGO aerogel/50% TiO ₂		5	55.9	127.8	47.6	118.4	~1.09	[127]
PW/rGO aerogel/80% TiO ₂		5	56.7	140.2	50.1	141.2	~1.16	[127]
PW/rGO aerogel/100% TiO ₂		5	55.7	127.8	50.4	128.5	1.22	[127]
PW/rGO aerogel/100% P25		5	55.9	110.2	47.6	109.1	0.94	[127]
PW		–	45.6	236.1	38.1	235.8	0.35	[128]

(continued on next page)

Table 3 (continued)

PCC	Preparation method	GO (mg/mL)	Melting Temperature (°C)	Melting Enthalpy (J/g)	Solidifying Temperature (°C)	Solidifying enthalpy (J/g)	Thermal Conductivity (W/m-K)	Ref.
	• Melting blending, vacuum impregnation							
PW/rGO aerogel		2	45.1	236.8	38.5	235.9	0.42	[128]
PW/rGO aerogel/SEBS ₅		2	45.2	226.3	40.5	226.1	0.43	[128]
PW/rGO aerogel/SEBS ₁₀		2	45.6	217.7	40.8	215.1	0.44	[128]
PW/rGO aerogel/SEBS ₁₅		2	44.9	193.9	40.6	192.7	0.42	[128]

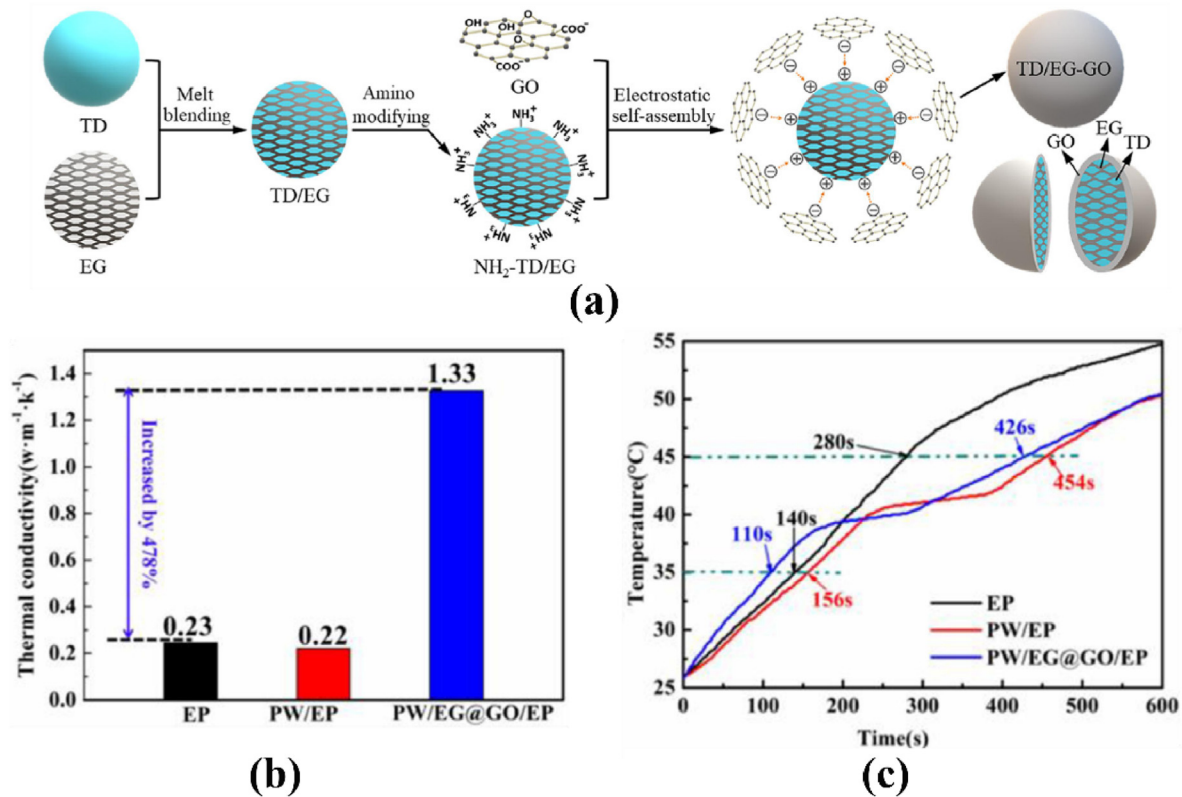


Fig. 9. (a) Schematic showing the preparation process of TD/EG-GO. Reproduced with permission [140]. Copyright 2021 Elsevier. (b) Comparison of the thermal conductivities of pure EP, PW/EP and PW/EG@GO/EP, and (c) Temperature-time curves of pure EP and epoxy resin composite PCMs filled with pure PW and PW/EG@GO. Reproduced with permission [141]. Copyright 2021 Elsevier.

formaldehyde and *n*-tetradecane as well as between CGMF and asphalt. The resulting core-shell phase change microcapsule is denoted here as CGMFPCM_x, where *x* is a number dependent on the graphene loading ratio in the composite shell. The performance of CGMFPCM_x was compared to a core-shell phase change microcapsule with a melamine formaldehyde shell without any graphene loading, denoted as MFPCM. When graphene loadings of 0.15% (CGMFPCM1) and 0.3% (CGMFPCM2) were used, the encapsulation ratio, defined as the ratio of the latent heat of the CGMFPCM_x to the latent heat of *n*-tetradecane, exceeded 44%, which was almost four times more than MFPCM. However, the encapsulation ratio decreased to 4% with 0.45% graphene loading (CGMFPCM3). Despite the low latent heat of the CGMFPCM3, it was able to maintain high thermal stability at an elevated temperature of 200 °C whereas significant mass losses were observed with CGMFPCM1 and CGMFPCM2. In addition, compared to MFPCM, both CGMFPCM2 and CGMFPCM3 increased the heat conductivity and heat storage

capacity by 1.55% and 5.7%, respectively. The incorporation of these phase change microcapsules in asphalt could thus reduce the temperature sensitivity of the modified asphalt, preventing it from undergoing severe deformations with seasonal temperature variations, and in doing so enhance its service lifetime.

Not only can graphene and its derivatives be added to organic-based shells for performance enhancement, but they can also be used in shells made of inorganic materials. Jiang et al. fabricated microencapsulated PCMs (MePCMs) comprising a PW PCM core and a shell made of calcium carbonate modified with GO [144]. A GO content of 1.0 wt% yielded a high encapsulation ratio of 73.2% and the leakage rate was reduced by 89.6%. The MePCM also had excellent thermal stability, improved mechanical properties, and a good thermal conductivity of 0.857 W/(m·K).

He et al. explored the possibility of using nanoencapsulation to enhance the phase change performance of sugar alcohols [145]. D-mannitol nanocapsules with a GO and silica composite shell were

synthesized using the GO-assisted sol-gel method. The melting and freezing latent heat values for the nanocapsules were 216.7 and 174.4 kJ/kg, respectively. The GO shell increased the thermal conductivity of the nanocapsules by 128.6% over pure D-mannitol PCM, but it did not significantly change the latent heat because the encapsulation by the GO/silica shell effectively protected the D-mannitol PCM from exposure to oxygen. With these thermal properties, the nanocapsules achieved an energy storage efficiency of 75.8% and a thermal reliability index of 96.1% after 50 thermal cycles.

Besides adding graphene and its derivatives into the shell of MePCMs, they may also be incorporated into the core and mixed with the PCM to modify the properties of the PCM itself. Chen et al. prepared MePCM where melamine formaldehyde was used as the shell material while the core PCM was an *in situ* polymerized mixture of GO-grafted octadecylamine (GO-ODA) and *n*-octadecane [146]. The addition of GO-ODA to *n*-octadecane promoted the crystallization of *n*-octadecane during polymerization, and this resulted in a decrease in the PCM's supercooling effect. Furthermore, the thermal conductivity of the MePCM improved by 38.5% with just a small amount (0.5 wt%) of GO-ODA added, demonstrating the superb effectiveness of the GO-ODA additive. At the GO-ODA loading of 0.5 wt%, the MePCM possessed a high latent heat of 207.2 J/g and an encapsulation efficiency of over 88.0%, where the encapsulation efficiency is defined as the ratio of the sum of the melting and freezing latent heats of MePCM to the sum of the melting and freezing latent heats of pure *n*-octadecane.

Graphene-based nanomaterials could also be added alongside MePCMs as a two-component mixture. In one instance, Wang et al. incorporated a two-component mixture of a commercial MePCM and graphene (0.2 wt%) into a wood material via vacuum impregnation to prepare a type of phase change energy storage wood (PCESW) [147]. With the porosity of the wood increased by delignification, the impregnation effect was enhanced as the relatively large MePCMs were able to enter the pores in the wood successfully and be distributed in the vessels. The resulting PCESW had excellent energy storage capacity, which could be adjusted by changing the MePCM content in the wood. The PCESW also possessed a suitable phase transition temperature for regulating indoor temperature. With the incorporation of graphene, the thermal conductivity of the PCESW was significantly improved, giving an enhancement of about 773% compared to a PCESW without any graphene additive.

Khezri et al. synthesized a MePCM comprising a core of PW and a shell of melamine formaldehyde formed by *in situ* polymerization [148]. To this MePCM, GnPs were added, physically mixed with the MePCM microcapsules, and dried to obtain a nanocomposite. GnPs were added for their role as a heat transfer promoter. Through the formation of this phase change nanocomposite, the leakage and low thermal conductivity of the PCM could be simultaneously overcome. Without having a significant influence on the composite's latent heat, the thermal conductivity and diffusivity of the MePCM were effectively enhanced in the composite by 48% and 93%, respectively, with 10 wt% of GnPs added. The composite was thermally stable up to 165 °C and possessed a latent heat of 95.97 J/g at a GnP loading of 1 wt%.

In summary, graphene-based nanomaterials can be incorporated into MePCMs in various ways to improve the MePCM's thermal and mechanical properties, among others. Firstly, graphene-based nanomaterials can be used as the shell material to encapsulate the PCM core, or they can also be included with another material (e.g., a polymer or ceramic) to create a composite shell material. Secondly, graphene-based nanomaterials may be included in the PCM core to improve the phase change behavior of the PCM itself. Finally, the graphene-based nanomaterial could be mixed with the MePCM as a separate component to form a MePCM-based PCC. In this case, MePCM microcapsules would form the disperse phase of the PCC whereas the graphene-based nanomaterial would be present in the continuous phase of this composite. This could allow the formation of an extensive thermally conductive network of the graphene-based nanomaterial within the continuous phase of the

PCC for more efficient heat transfer. The improved heat dissipation and temperature-regulating properties of graphene-based microencapsulated PCMs would be a desirable material to be integrated into textiles to develop smart clothing or even in electronic devices whereby these materials can extend the lifespan and efficiency of these devices.

5. Applications of graphene-based PCCs

5.1. Solar energy harvesting systems

Among existing energy storage forms, thermal energy storage from solar energy is probably the most actively and extensively researched. Solar energy is a renewable energy source, environmentally friendly, and abundant. However, because there is a constant mismatch between solar energy demand and supply, efficient and reliable energy storage and conversion systems are needed to reduce this demand and supply gap. In this regard, graphene-based PCCs with high thermal energy storage densities are attractive for their implementation in solar thermal systems to improve their overall reliability, efficiency and performance.

Using magnesium nitrate hexahydrate, carboxymethyl cellulose, and graphene, Wang et al. prepared solar-driven PCCs with good energy storage and photo-thermal conversion performance [149]. To overcome the phase separation issue for inorganic PCMs, carboxymethyl cellulose was added as a thickening agent while the addition of 5 wt% of graphene increased the thermal conductivity from 0.34 W/(m·K) to 0.99 W/(m·K), an enhancement of 191.18%. The addition of graphene did not significantly affect the latent heat of the PCCs, keeping it at an acceptable value of 122.8 kJ/kg. Furthermore, due to graphene's outstanding optical absorption capacity over the wavelengths ranging from 325 nm to 1400 nm and 1570 nm–1860 nm and its photothermal conversion property, the thermal absorption capacity of the PCCs increased from 46.54% to 79.12% and the photo-thermal conversion efficiency reached 69.73%, respectively. Similarly, Cui et al. prepared a low-cost and stable PCC for solar thermal utilization by using a salt hydrate (sodium acetate trihydrate) (SAT) as the PCM with added xanthan gum to prevent phase segregation [150]. To solve the supercooling issue and to enhance the photothermal charging efficiency, GO@SiO₂ was fabricated with the intention of adding it to SAT as a nucleation agent to suppress its supercooling. At the same time, silicon carbide foam was incorporated to enhance the thermal conductivity of the PCC. With the addition of 1.0 wt % GO@SiO₂, the supercooling degree of SAT was drastically reduced from 32.0 °C to 0.5 °C, and the solar energy charging efficiency was increased from 60.1% to 79.9%. However, it was noted that the addition of 2.0 wt% GO@SiO₂ increased the supercooling degree to 1.4 °C due to aggregation, which will lead to an increase in the specific surface area and the reduced effective contact area between SAT and GO@SiO₂. The introduction of the silicon carbide foam further increased the solar energy charging efficiency to 92.5%. With both GO@SiO₂ and silicon carbide foam, the PCC had a latent heat and thermal conductivity of 193.2 J/g and 1.65 W/(m·K) respectively.

Lu et al. fabricated a PCC for photothermal conversion using PEG-based PU as the PCM, whereas GO and wood powder (WP) was used as a thermally conductive filler as well as the supporting matrix for the PCM [151]. As mentioned in the preceding sections, the abundance of oxygen-containing groups on GO facilitates dispersion, and therefore in this work, GO was embedded on the surface of hydrophilic WP via strong interfacial interactions to create the WP@GO composite filler. With a 4 wt% GO in WP@GO and a 20 wt% of WP@GO in the PCC, the PCC achieved a thermal conductivity of 1.87 W/(m·K) and a good latent heat value of 140.2 J/g. The effect that GO had on the PCC was evident in a light-to-thermal conversion study, whereby the PCC with GO (SSPCM-4) demonstrated faster heating and cooling rates compared to the PCC without GO (SSPCM-1) (Fig. 10).

In another work, Yang et al. introduced scaffolds made of BN micro sheets bridged with rGO to form a thermally conductive scaffold for the encapsulation of PEG PCM [152]. In addition to its thermal conductivity

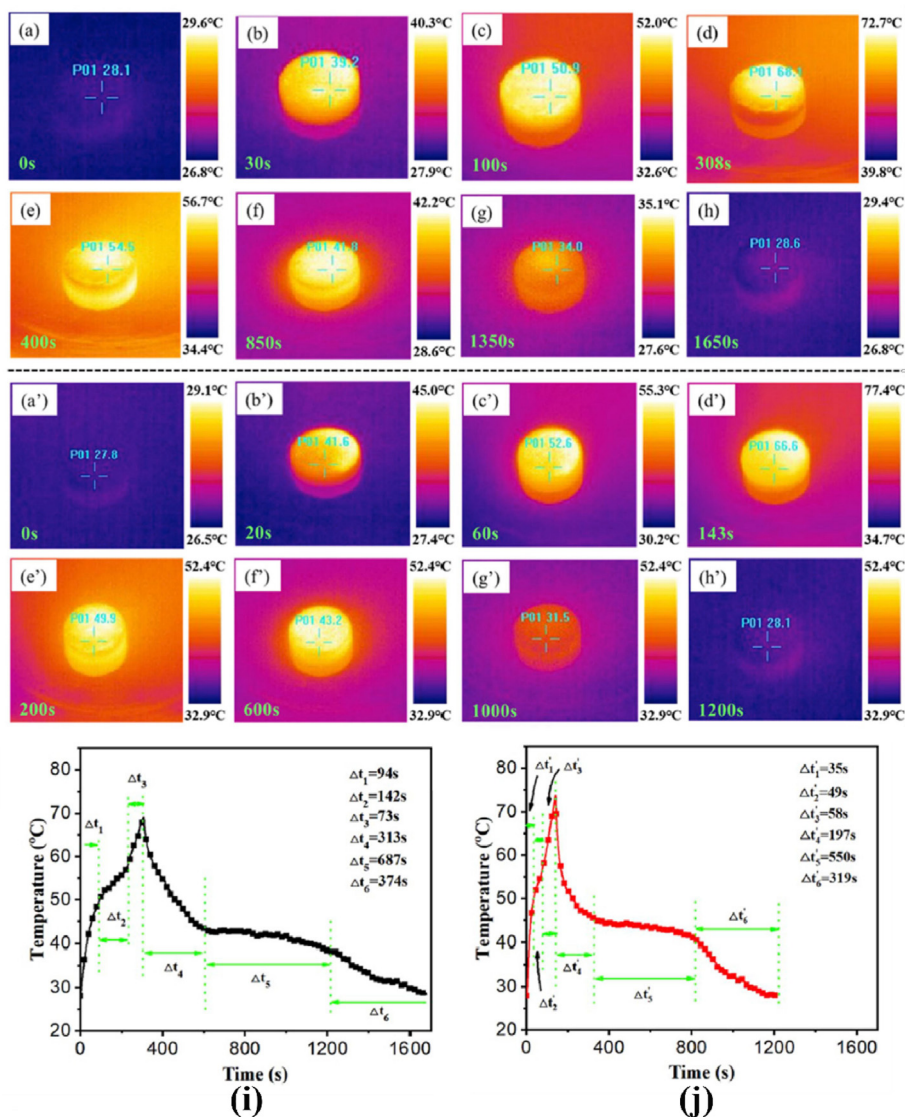


Fig. 10. Representative thermal infrared images about the light-thermal conversion performance for (a–h) SSPCM-1, (a'–h') SSPCM-4 with different time, Temperature-time curves for (i) SSPCM-1 and (j) SSPCM-4 during the light-thermal conversion process. Reproduced with permission [151]. Copyright 2021 Elsevier.

and shape stability, the rGO/BN scaffold (denoted as rGBP) also exhibited good electro-thermal/photo-thermal energy conversion. To obtain rGO/BN, BN was first bridged with GO sheets by freeze-drying to form GO/BN (denoted as GBP). Subsequently, microwave irradiation was employed for the *in situ* reduction of graphene oxide in GBP to rGBP. Since BN is transparent to microwaves, GO was able to absorb most of the microwave irradiation for its reduction to rGO. Comparing rGBP to GBP and GBP with pretreatment (pGBP), rGBP exhibited the highest heating and cooling rate (Fig. 11a and b), which attests to its enhanced thermal conductivity of 1.06 W/(m·K) with 14.4 wt% BN. The electro-thermal and photo-thermal conversion efficiencies of rGBP were also high, with values of up to 87.9% and 87.4%, respectively. The electro-thermal study indicated that a sudden increase in temperature in the initial stage occurred when a constant voltage of 7 V was applied to rGBP, which demonstrated the quick response time for the electro-thermal effect to kick in (Fig. 11c). In addition, both rGBP and GBP demonstrated fast heating rates under solar illumination owing to the addition of rGO as an effective photo-thermal conversion filler (Fig. 11d).

Meanwhile, using a combination of circular impregnation and *in situ* reduction process, Tao et al. employed a facile process to fabricate multifunctional 3D continuous sponge frameworks (SF) wrapped with rGO and impregnated with PW PCM [153]. The SF was first immersed in

an aqueous suspension of GO to ensure that the SF was well-coated with the GO nanoflakes before the *in situ* reduction process to rGO was carried out in hydroiodic acid (HI). The rGO@SF was then infused with PW by vacuum impregnation to form the PCC of rGO@SF/PW. It was found that the rGO@SF/PW PCC had a phase change enthalpy of 170.4 J/g and a solar-thermal conversion efficiency of 85%. Compared to pristine PW, the thermal conductivity of rGO-PW only increased by 20% whereas an enhancement of 84% was noted for rGO@SF PW PCC. This demonstrated that having a 3D continuous framework structure is crucial as the continuous porous structure can provide an effective channel for phonon transfer.

5.2. Building materials for thermal regulation

The enhanced thermal properties of graphene-based PCCs, such as high thermal energy storage capacity, heat transfer efficiency, and thermal conductivity, make them promising candidates for energy management in buildings. Graphene-based PCCs can be integrated into different building materials like cement, coatings, or insulation materials to decrease heating and cooling loads, regulate indoor temperature and therefore reduce energy consumption. Furthermore, the outstanding mechanical properties of graphene can confer improved longevity and

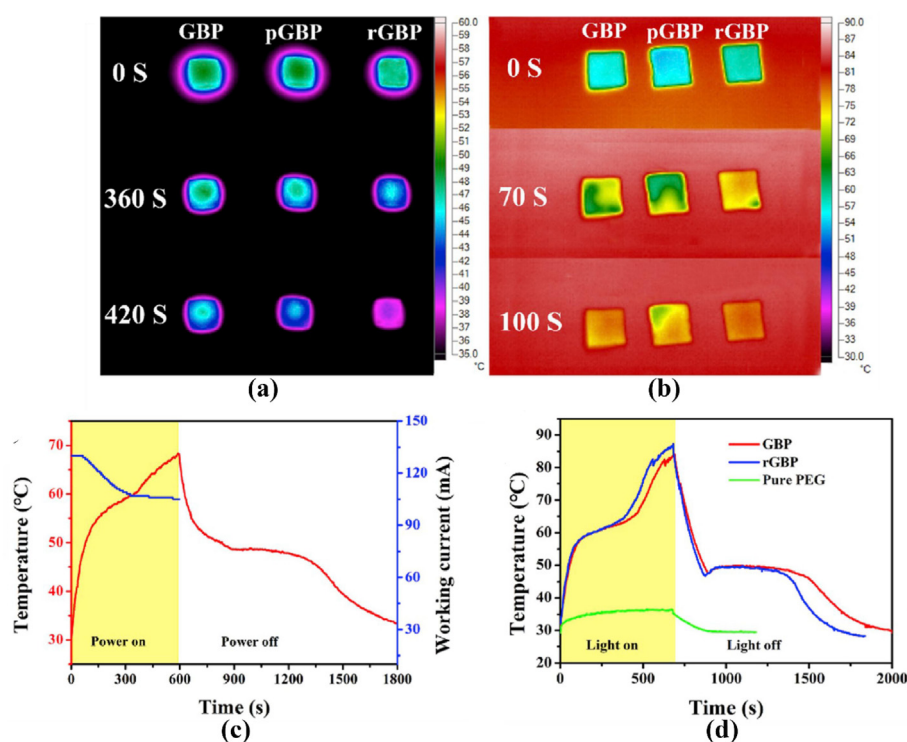


Fig. 11. Thermal transport evolution of different composite PCMs during (a) cooling and (b) heating processes. (c) Temperature-time curve and current-time curve during electro-thermal conversion process of rGBP, and (d) Temperature-time curves during photo-thermal conversion process of pure PEG, GBP and rGBP. Reproduced with permission [152]. Copyright 2020 Elsevier.

durability to the overall building components.

To investigate the performance of MePCMs in gypsum materials, Zhang et al. studied the effect of different contents of MePCMs in gypsum, as well as the effect of adding GO as an emulsifier, along with two other food-grade emulsifiers of Span80 and Tween 80, for the *in situ* polymerization of the MePCM [154]. Paraffin was used as the core which was encapsulated by a melamine-urea-formaldehyde (MUF) resin shell. The emulsifiers were first added into the liquid paraffin to make an emulsion, after which it was mixed with aqueous MUF for *in situ* polymerization to take place, forming the GO-containing Paraffin@MUF MePCM. The MePCM had a latent heat of 32.2 J/g and an encapsulation ratio was 49.9%. It was noted that GO aided in the uniformity of the particle sizes, thus demonstrating that the addition of an emulsifier had a positive influence on particle distribution. The MePCM was then added to gypsum powder. As the content of MePCM in the gypsum increased, the thermal conductivity decreased because the MePCM has a lower thermal conductivity value compared to gypsum (Fig. 12a). Similarly, the flexural and compressive strength of the gypsum composite decreased with increasing MePCM content (Fig. 12b) and above 30% of MePCM, agglomeration of MePCMs occurred resulting in low compatibility. Nevertheless, the incorporation of MePCMs in the gypsum materials displayed a better thermal insulation performance that is advantageous for temperature regulation in buildings. As compared to unmodified gypsum which took 1355 s to reach 40 °C, the gypsum composites containing 20%, 30%, and 50% MePCM took significantly longer times of 1490 s, 1700 s, and 1840 s respectively to reach the same temperature.

Meanwhile, Jin et al. prepared PCCs consisting of lauric-stearic acid/treated diatomite/graphene (LA-SA/Dt-GR) and investigated their thermal and mechanical performances in mortar [155]. SEM analysis revealed that LA-SA was successfully incorporated into the porous structure of Dt and the PCC maintained good thermal stability after 1000 thermal cycles. The addition of graphene not only increased the thermal conductivity of the phase change mortar, with a 2 wt% content of graphene enhancing the thermal conductivity by 274% (Fig. 12c), but also

improved the melting and cooling efficiencies of the PCCs significantly. Unsurprisingly, the mechanical (flexural and compression) strength of the phase change mortar decreased with increasing content of the mechanically softer PCCs, and this did not significantly change with a variation in the added graphene content (Fig. 12d). In another largely similar work, the same authors prepared capric-stearic acid/montmorillonite (CA-SA/MMT) PCCs with added graphene fillers and incorporated them into mortar as well [156]. MMT was used to prevent the leakage of the CA-SA PCM and a CA-SA to MMT mass ratio of 35:65 effectively did so, while also providing good thermal stability even after 300 thermal cycles. The addition of 2 wt% graphene served to increase the thermal conductivity of the PCC by 156.4%. Similar to their previous work conducted, a 10 wt% of the graphene-containing CA-SA/MMT PCC in the phase change mortar displayed weakened mechanical strength, with a decrease in flexural and compression strength by 43.8% and 66.2% respectively. Nevertheless, the suitable thermal properties and good chemical stabilities of these PCCs still make them attractive materials to be used for TES applications in buildings.

On the other hand, a mesoporous PCC that can be added as an additive to Portland cement composites was designed by Shamsaei et al. [157]. Using an *in situ* dissolution-coprecipitation strategy, SiO₂ nanoparticles were first deposited on both sides of GO nanosheets, from which a vertically interconnected network of 2D calcium silicate hydrate (CSH) was grown, resulting in a sandwiched CSH-GO-CSH (CGC) structure. Thereafter, the CGC structure was infiltrated with the PCM of LA to produce a mesoporous LA@CGC PCC. LA@CGC possessed a high specific surface area (677 m²/g), large pore volume (~2.5 cm³/g), and a high latent heat value of 118.0–127.6 J/g that was stable after 50 thermal cycles. Compatibility studies for LA@CGC in Portland cement were conducted, and it was revealed that at 1–5 wt% of loading, LA@CGC were uniformly distributed within the cement material and could accelerate the hydration reaction of the cement as well. It was suggested that CSH in LA@CGC can promote the nucleation of CSH products and accelerate the strength development of Portland cement. The results

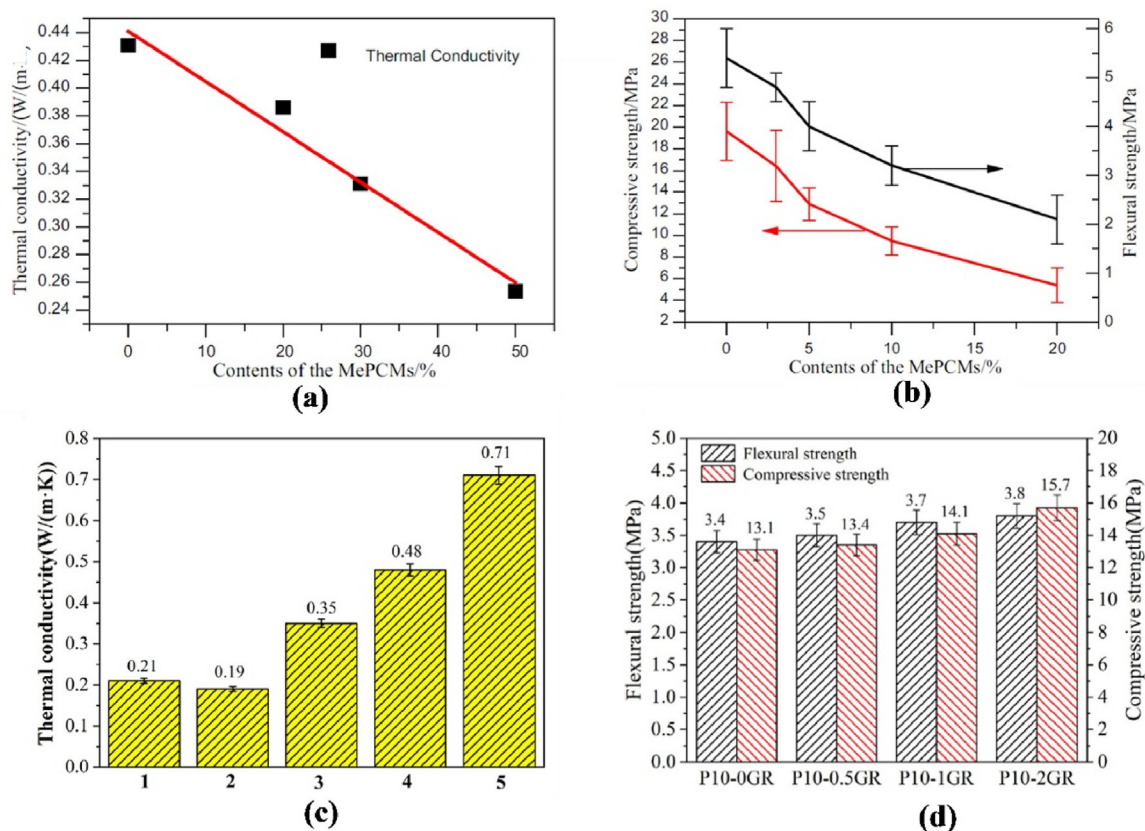


Fig. 12. (a) Thermal conductivity of gypsum building materials incorporated with different contents of the GO-containing paraffin@MUF MePCMs, (b) Relationship of the mechanical strength and the gypsum-based materials incorporated with different contents of GO-containing paraffin@MUF MePCMs. Reproduced with permission [154]. Copyright 2019 Elsevier. (c) Thermal conductivities of PCMs with different contents of treated diatomite (Dt) and graphene (GR): 1) LA-SA; 2) LA-SA/Dt; 3) LA-SA/Dt/0.5 wt%GR; 4) LA-SA/Dt/1 wt%GR; 5) LA-SA/Dt/2 wt%GR, (d) Flexural and compressive strengths of phase change mortar containing 10 wt% LA-SA/Dt/GR composite PCMs with different percentages of GR. Reproduced with permission [155]. Copyright 2021 Elsevier.

indicated that LA@CGC could be added to cement not only as a thermal storage material but also as an additive to fabricate cement with slow hardening and setting behavior.

5.3. Battery thermal management

Batteries are used in a myriad of applications, such as electric vehicles [158], renewable energy systems [159,160], and portable electronics [161,162]. During the continuous process of charging and discharging, a large amount of heat is generated and slow dissipation of this heat will result in a deteriorative performance, shortened service span, and even pose serious safety risks. Therefore, the thermal management of batteries is crucial for ensuring battery safety, prolonging its lifespan, maximizing its performance, and maintaining its efficiency. In this regard, PCMs serve as a passive and effective solution via heat absorption, dissipation, and thermal regulation.

Wang et al. designed a phase change system to achieve efficient heat storage and controllable heat release for Li-ion batteries [163]. To improve the thermal properties of the PCM calcium chloride hexahydrate, the use of MOF-derived carbon (MOF-C)/GO hybrid aerogel was proposed as a filler for the PCM to make a PCC. The synergistic effect between MOF-C and GO, combining their high specific surface area and surface-active sites to interact with the PCM, could enhance the PCM loading content and reduce PCM leakage, while the formation of a hybrid aerogel 3D network improved the thermal conductivity of the PCC. By gradually varying the added content of strontium chloride hexahydrate (a nucleating agent) along the thickness of the hybrid aerogel network to effectively form a multi-layered PCM having a different nucleating agent content in each layer, a cascade phase change mechanism could be

achieved due to the creation of a gradient of multiple freezing points across the aerogel network, which helps to retard the release of heat from the battery. In this way, the discharge capacity of the battery was successfully enhanced under a cold environment, and the warming time of the battery was prolonged when compared with a single-layer PCM with only one freezing point. With the cascade hybrid MOF-C/GO PCC, a remarkable heat regulation performance was realized, which included a high latent heat (140 J/g), increased thermal conductivity (1.36 W/(m·K)), shape stability (at 10 °C above the melting point), and durability after 100 times of thermal cycling. A battery test conducted to compare the performance of a single-layer PCM and a two-layer PCM (two freezing points) revealed that with an additional freezing point, the heat-releasing time of the latter was extended by 200%, its discharge capacity increased by 50%, and the low-temperature resistance of the battery was reduced.

Using the structure of spider webs as an inspiration, Lin et al. fabricated PCCs containing a graphene skeleton (GS) with a 3D spider web (sw)-like structure in a PCM of PW [164]. Using a combination of methods such as hydrothermal treatment, radial freeze-casting, and vacuum impregnation, the sw-GS/PW PCC exhibited good thermal stability and shape stability due to the rigid structure of the graphene skeleton. Even with a low GO content of 2.25 vol%, high cross-plane and in-plane thermal conductivities of 2.58 and 1.78 W/(m·K) were reported, corresponding to enhancements of ~1260% and ~840%, respectively. A thermal management application study was conducted whereby the sw-GS/PW PCC was wrapped around a battery. The PCC contributed to significantly reducing the temperature rise during the continuous operation of the battery. Similarly, Goli et al. used single-layer graphene and few-layer graphene and mixed them with PW to investigate their thermal

performance with Li-ion batteries [165]. The thermal conductivity of the hybrid PCM increased by more than two orders of magnitude whereas the latent heat storage capability remained unchanged. Simulation studies also revealed that the temperature rise inside the battery pack could be significantly reduced and the temperature profile was more uniform as well.

5.4. Other applications

Besides the aforementioned applications of graphene-based PCCs, several different types of applications of these materials have been investigated as well. With the goal of fabricating highly heat-dissipating textiles for cool clothing, Ruiz-Calleja et al. investigated the differences in the thermal behavior of a cellulosic fabric by applying coating pastes containing PCM only, containing graphene only, as well as containing a combination of these two materials [166]. Both graphene-only and PCM-only coatings exhibited similar behavior when dissipating heat, due to the excellent thermal conductivity of graphene and the PCM's ability to store latent heat. By combining both materials in the same coating, the synergy between the two materials achieved a higher dissipation of thermal energy during the heating process, and the incorporation of graphene also enhanced the heat storage capacity of the PCM. In light of this finding, further studies on other textile substrates such as those made with hollow fibers for better thermal properties tunability or using different methods of incorporating these PCC materials into textiles could be carried out as well.

Lin et al. fabricated thermally induced flexible wood via the impregnation of delignified wood (DW) with a PCC of modified PEG and graphene [167]. By removing lignin, the DW had a more porous structure compared to pure wood which facilitated the PCC impregnation process. In addition, the formation of multi-channeled and crumpled cell structures with numerous cracks in DW led to an elastic and more flexible structure. The PCCs with added graphene showed good thermal stability and high latent heat while the DW impregnated with the PCC with 3 wt% graphene (TESWG3) increased the thermal conductivity by about 414% compared to pure DW without PCC. A flexibility test revealed the unique shape memory ability of TSWG3 which was soft and flexible when heated to 35 °C and it became stiff when cooled to 20 °C. This was attributed partially to the increase in hydrogen bonding between the cellulose chains which improved the compressive and tensile strength of DW, and partially to the intrinsic flexibility of the PCC material in TSWG3. Potential applications for the PCC-impregnated DW are in furniture or flooring that could contribute to temperature regulation in buildings.

Cold chain transportation is essential for the transportation of temperature-sensitive items such as fresh produce and vaccines. It requires a large amount of energy that primarily comes from the vapor compression refrigeration cycle supplied by diesel engines to keep the products cool. To improve energy efficiency and reduce emissions, PCMs can be integrated into conventional refrigeration technologies. Nie et al. investigated the performance of a portable box for cold chain distribution by incorporating a PCC of paraffin PCM, graphene, and fumed silica into it [168]. To address the issue of leakage of the PCM, fumed silica was added to aid in the charging and discharging process of the PCM by improving the nucleation process. The PCC displayed good chemical stability and thermal reliability while a loading of 4 wt% silica was sufficient to prevent PCM leakage. It was proposed that the interaction between the silica and paraffin particles was strengthened during the impregnation process and the resulting compact colloidal-like structure of the PCC prevented the phase segregation and leakage when the PCM is in the liquid state. The incorporation of 1 wt% graphene significantly increased the thermal conductivity of the composite by 55.4% as well. More importantly, a temperature within the range of 2–8 °C could be maintained in the modified portable cool box for up to 11.5 h, with improvements in the overall energy and charging efficiencies by 12.58% and 6.09%, respectively, compared to an unmodified cool box.

To manage the heat produced by electronic components, PCMs and heat pipes can be added to enhance the cooling efficiency. Heat pipes aid in heat dissipation while PCMs help to reduce the temperature of the heat sink. To further increase the efficiency of heat sinks, Ali utilized a heat pipe, along with PW as a PCM and GO nanoparticle fillers to be mixed with the PCM [169]. The performance of the developed PCC with the heat pipe was studied using different heating loads (1 kW/m², 1.5 kW/m², and 2.5 kW/m²). At lower heating loads of 1 kW/m² and 1.5 kW/m², the heat sinks modified with the PW/GO PCC and heat pipe showed temperature reduction of up to 34.06% and 36.45%, respectively for each heating load. At a higher heating load of 2.5 K kW/m², a temperature reduction of 37.54% was observed with the PW/GO PCC and heat pipe, while the heat sink aided with PW, heat pipe, and a fan had a better temperature reduction of 42.81%. The better cooling efficiency by the fan than the GO nanoparticles is due to the settling of the GO nanoparticles within the PW, causing agglomeration and thus the overall thermal conductivity was not improved to a large extent.

In summary, we have covered some of the more popular applications and some more unconventional ones that could benefit from graphene-based PCCs for thermal energy management and storage. PCCs play an important role especially in the broad domains of energy sustainability, where they are greatly desired for applications in solar energy, building materials and batteries. Other applications where thermal storage and management can be used include cold chain logistics and high power electronic devices and more which are not mentioned here. The diversity of applications where PCCs can be applied highlights the importance of continuous innovation in heat storage and regulation technology. Based on ongoing research trends and the promising characteristics of graphene-based PCCs, they can potentially be applied in a wider range of applications. For example, they could be employed in grid-scale energy storage applications to better integrate renewable energy and enhance energy efficiency in the power grid. Applications in high-performance computing systems and data centers to address thermal management challenges and reduce cooling costs could be a possibility too. Nevertheless, collaborative research and sustained efforts are required to overcome any economic, technical, and scalability issues if such practical applications were to come to fruition.

6. Concluding remarks

The present article provides an overview of the latest advancements in the implementation of graphene-based PCCs in TES applications. There exist several methodologies for the preparation of graphene-based polymer composite materials, including but not limited to melt mixing, vacuum filtration, ultrasonication, in-situ synthesis, electrospinning, solution casting, and layer-by-layer assembly. The fabrication of graphene-based PCCs with desired properties can be achieved through the utilization of these methods either individually or in conjunction with others. The principal benefit of graphene and its derivatives lies in their elevated thermal conductivity, which imparts to PCMs a swifter and more effective mode of heat transfer. Consequently, these composite materials exhibit favorable characteristics for a diverse array of thermal management applications, including but not limited to building infrastructure, energy storage technology, and electronic devices. Incorporating graphene and its derivatives into PCMs has been found to enhance their mechanical properties, including flexibility and strength in a synergistic manner. This, in turn, leads to the PCM's increased durability and reduced susceptibility to damage or leakage. The phase change enthalpy is the main characteristic of PCM systems and yet little/no in-depth comparative study has been carried out to investigate the interactions between PCMs and graphene-based nanostructures. As such, more research studies in this area is essential for the development of better graphene-based PCCs.

Despite their many advantages, graphene and its derivatives have several limitations and challenges that need to be addressed. The cost of producing high-quality graphene is still relatively high compared to

other materials due to its energy-intensive and complex synthesis processes. Additionally, scalability poses a challenge for graphene production. Graphene and its derivatives tend to form agglomerates which can limit their performance and they are susceptible to degradation and oxidation under certain environmental conditions. Furthermore, research has indicated potential toxicity issues associated with the utilization of specific graphene derivatives, including GO, thereby restricting its application in certain contexts. Therefore, the future outlook of graphene-based PCCs will depend on the aforementioned issues such as cost reductions and scalability, in addition to the optimization of the performance of these materials and the advancement in manufacturing techniques.

Still, based on their enhanced thermal performance, graphene-based PCCs are distinctly attractive over conventional PCMs, indicating a promising outlook for their application. Moreover, there exists a possibility for these systems to exhibit eco-friendliness by utilizing sustainable and biodegradable PCMs like fatty acids, in lieu of their petroleum-based counterparts. Despite being in the nascent stages of development, graphene-based PCCs have exhibited considerable potential in diverse applications, as evidenced by the research efforts as discussed in this review. Further investigations and advancements in this domain to tackle the existing obstacles and constraints may potentially culminate in the commercialization of graphene-based PCCs in the imminent future.

Declaration of competing interest

Xian Jun Loh is an editorial board member for [Nano Materials Science] and was not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

Acknowledgments

The authors acknowledge the support from Grant No. 2022VBA0023 funded by the Chinese Academy of Sciences President's International Fellowship Initiative.

References

- [1] S. Wang, J.K. Muiruri, X.Y.D. Soo, S. Liu, W. Thitsartam, B.H. Tan, A. Suwardi, Z. Li, Q. Zhu, X.J. Loh, Bio-polypropylene and polypropylene-based biocomposites: solutions for a sustainable future, *Chem. Asian J.* (2022) e202200972, <https://doi.org/10.1002/asia.202200972>.
- [2] J.K. Muiruri, J.C.C. Yeo, X.Y.D. Soo, S. Wang, H. Liu, J. Kong, J. Cao, B.H. Tan, A. Suwardi, Z. Li, J. Xu, X.J. Loh, Q. Zhu, Recent advances of sustainable short-chain length polyhydroxyalkanoates (ScL-PHAs)–Plant biomass composites, *Eur. Polym. J.* (2023) 111882, <https://doi.org/10.1016/j.eurpolymj.2023.111882>.
- [3] Z. Li, X.J. Loh, Recent advances of using polyhydroxyalkanoate-based nanovehicles as therapeutic delivery carriers, *Wiley Interdiscip. Rev. Nanomed. Nanobiotechnol.* 9 (3) (2017) e1429, <https://doi.org/10.1002/wnan.1429>.
- [4] M.H. Chua, K.L.O. Chin, X.J. Loh, Q. Zhu, J. Xu, Aggregation-induced emission-active nanostructures: beyond biomedical applications, *ACS Nano* 17 (3) (2023) 1845–1878, <https://doi.org/10.1021/acsnano.2c10826>.
- [5] M. Wu, S. Wu, Y. Cai, R. Wang, T. Li, Form-stable phase change composites: preparation, performance, and applications for thermal energy conversion, storage and management, *Energy Storage Mater.* 42 (2021) 380–417, <https://doi.org/10.1016/j.ensm.2021.07.019>.
- [6] Q. Zhu, M.H. Chua, P.J. Ong, J.J.C. Lee, K.L.O. Chin, S. Wang, D. Kai, R. Ji, J. Kong, Z. Dong, Recent advances in nanotechnology-based functional coatings for the built environment, *Mater. Today Adv.* 15 (2022) 100270, <https://doi.org/10.1016/j.mtadv.2022.100270>.
- [7] P.J. Ong, Y.Y. Lum, X.Y.D. Soo, S. Wang, P. Wang, D. Chi, H. Liu, D. Kai, C.-L.K. Lee, Q. Yan, J. Xu, X.J. Loh, Q. Zhu, Integration of phase change material and thermal insulation material as a passive strategy for building cooling in the tropics, *Construct. Build. Mater.* 386 (2023) 131583, <https://doi.org/10.1016/j.conbuildmat.2023.131583>.
- [8] P.J. Ong, Y. Leow, X.Y.D. Soo, M.H. Chua, X. Ni, A. Suwardi, C.K.I. Tan, R. Zheng, F. Wei, J. Xu, X.J. Loh, D. Kai, Q. Zhu, Valorization of Spent coffee Grounds: a sustainable resource for Bio-based phase change materials for thermal energy storage, *Waste Manage. (Tucson, Ariz.)* 157 (2023) 339–347, <https://doi.org/10.1016/j.wasman.2022.12.039>.
- [9] N.H. Solangi, N.M. Mubarak, R.R. Karri, S.A. Mazari, A.S. Jatoti, J.R. Koduru, M.H. Dehghani, MXene-based phase change materials for solar thermal energy storage, *Energy Convers. Manag.* 273 (2022) 116432, <https://doi.org/10.1016/j.enconman.2022.116432>.
- [10] T. Li, M. Wu, S. Wu, S. Xiang, J. Xu, J. Chao, T. Yan, T. Deng, R. Wang, Highly conductive phase change composites enabled by vertically-aligned reticulated graphite nanoplatelets for high-temperature solar photo/electro-thermal energy conversion, harvesting and storage, *Nano Energy* 89 (2021) 106338, <https://doi.org/10.1016/j.nanoen.2021.106338>.
- [11] E. Yildirim, Q. Zhu, G. Wu, T.L. Tan, J. Xu, S.-W. Yang, Self-organization of PEDOT: PSS induced by green and water-soluble organic molecules, *J. Phys. Chem. C* 123 (15) (2019) 9745–9755, <https://doi.org/10.1021/acs.jpcc.9b01716>.
- [12] H. Zhou, M.H. Chua, Q. Zhu, J. Xu, High-performance PEDOT: PSS-based thermoelectric composites, *Compos. Commun.* 27 (2021) 100877, <https://doi.org/10.1016/j.coco.2021.100877>.
- [13] X. Yong, G. Wu, W. Shi, Z.M. Wong, T. Deng, Q. Zhu, X. Yang, J.-S. Wang, J. Xu, S.-W. Yang, Theoretical search for high-performance thermoelectric donor–acceptor copolymers: the role of super-exchange couplings, *J. Mater. Chem. A* 8 (41) (2020) 21852–21861, <https://doi.org/10.1039/D0TA05765G>.
- [14] T. Tang, A.K.K. Kyaw, Q. Zhu, J. Xu, Water-dispersible conducting polyazulene and its application in thermoelectrics, *Chem. Commun.* 56 (65) (2020) 9388–9391, <https://doi.org/10.1039/D0CC03840G>.
- [15] V. Skurkyte-Papieviene, A. Abraitiene, A. Sankauskaite, V. Rubeziene, J. Baltusnikaite-Guzaitiene, Enhancement of the thermal performance of the paraffin-based microcapsules intended for textile applications, *Polymers* 13 (7) (2021) 1120, <https://doi.org/10.3390/polym13071120>.
- [16] Y. Lu, X. Xiao, J. Fu, C. Huan, S. Qi, Y. Zhan, Y. Zhu, G. Xu, Novel smart textile with phase change materials encapsulated core-sheath structure fabricated by coaxial electrospinning, *Chem. Eng. J.* 355 (2019) 532–539, <https://doi.org/10.1016/j.cej.2018.08.189>.
- [17] Y. Liao, J. Li, S. Li, X. Yang, Super-elastic and shape-stable solid-solid phase change materials for thermal management of electronics, *J. Energy Storage* 52 (2022) 104751, <https://doi.org/10.1016/j.est.2022.104751>.
- [18] W.-Y. Wu, G.M.D. Yeo, S. Wang, Z. Liu, X.J. Loh, Q. Zhu, Recent Progress in Polyethylene-enhanced Organic Phase Change Materials Composite for Energy Management, *Chem. Asian J.* e202300391, <https://doi.org/10.1002/asia.202300391>.
- [19] S.Y. Kee, Y. Munusamy, K.S. Ong, Review of solar water heaters incorporating solid-liquid organic phase change materials as thermal storage, *Appl. Therm. Eng.* 131 (2018) 455–471, <https://doi.org/10.1016/j.applthermaleng.2017.12.032>.
- [20] H. Ke, Y. Li, J. Wang, B. Peng, Y. Cai, Q. Wei, Ag-coated polyurethane fibers membranes absorbed with quinary fatty acid eutectics solid-liquid phase change materials for storage and retrieval of thermal energy, *Renew. Energy* 99 (2016) 1–9, <https://doi.org/10.1016/j.renene.2016.06.033>.
- [21] X.Y. Soo, J.K. Muiruri, J.C. Yeo, Z.M. Png, A. Sng, H. Xie, R. Ji, S. Wang, H. Liu, J. Xu, X.J. Loh, Q. Yan, Z. Li, Q. Zhu, Polyethylene glycol/poly(lactic acid) block copolymers as solid–solid phase change materials, *SmartMat* (2023) e1188, <https://doi.org/10.1002/smm2.1188>.
- [22] X.Y.D. Soo, Z.M. Png, M.H. Chua, J.C.C. Yeo, P.J. Ong, S. Wang, X. Wang, A. Suwardi, J. Cao, Y. Chen, A highly flexible form-stable silicone-octadecane PCM composite for heat harvesting, *Mater. Today Adv.* 14 (2022) 100227, <https://doi.org/10.1016/j.mtadv.2022.100227>.
- [23] X.Y.D. Soo, Z.M. Png, X. Wang, M.H. Chua, P.J. Ong, S. Wang, Z. Li, D. Chi, J. Xu, X.J. Loh, Q. Zhu, Rapid UV-curable form-stable polyethylene-glycol-based phase change material, *ACS Appl. Polym. Mater.* 4 (4) (2022) 2747–2756, <https://doi.org/10.1021/acsaapm.2c00059>.
- [24] S. Wu, T. Li, Z. Tong, J. Chao, T. Zhai, J. Xu, T. Yan, M. Wu, Z. Xu, H. Bao, High-performance thermally conductive phase change composites by large-size oriented graphite sheets for scalable thermal energy harvesting, *Adv. Mater.* 31 (49) (2019) 1905099, <https://doi.org/10.1002/adma.201905099>.
- [25] T. Li, J. Xu, D. Wu, F. He, R. Wang, High energy-density and power-density thermal storage prototype with hydrated salt for hot water and space heating, *Appl. Energy* 248 (2019) 406–414, <https://doi.org/10.1016/j.apenergy.2019.04.114>.
- [26] S. Wu, T. Li, M. Wu, J. Xu, Y. Hu, J. Chao, T. Yan, R. Wang, Highly thermally conductive and flexible phase change composites enabled by polymer/graphite nanoplatelet-based dual networks for efficient thermal management, *J. Mater. Chem. A* 8 (38) (2020) 20011–20020, <https://doi.org/10.1039/D0TA05904H>.
- [27] A. Yadav, A. Verma, A. Kumar, H. Dashmana, A. Kumar, P. Bhatnagar, V. Jain, Recent advances on enhanced thermal conduction in phase change materials using carbon nanomaterials, *J. Energy Storage* 43 (2021) 103173, <https://doi.org/10.1016/j.est.2021.103173>.
- [28] Z.A. Qureshi, H.M. Ali, S. Khushnood, Recent advances on thermal conductivity enhancement of phase change materials for energy storage system: a review, *Int. J. Heat Mass Tran.* 127 (2018) 838–856, <https://doi.org/10.1016/j.jheatmasstransfer.2018.08.049>.
- [29] P.J. Ong, Z.X.J. Heng, Z. Xing, H.Y.Y. Ko, P. Wang, H. Liu, R. Ji, X. Wang, B.H. Tan, Z. Li, J.W. Xu, X.J. Loh, E. Ye, Q. Zhu, Wax from pyrolysis of waste plastics as a potential source of phase change material for thermal energy storage, *Trans. Tianjin Univ.* (2022) 1–10, <https://doi.org/10.1007/s12209-022-00346-7>.
- [30] Y. Zhang, Z. Jia, A.M. Hai, S. Zhang, B. Tang, Shape-stabilization micromechanisms of form-stable phase change materials-A review, *Compos. - A: Appl. Sci. Manuf.* (2022) 107047, <https://doi.org/10.1016/j.compositesa.2022.107047>.
- [31] Z.M. Png, X.Y.D. Soo, M.H. Chua, P.J. Ong, J. Xu, Q. Zhu, Triazine derivatives as organic phase change materials with inherently low flammability, *J. Mater. Chem. A* 10 (7) (2022) 3633–3641, <https://doi.org/10.1039/D1TA07422A>.

- [32] A. Olabi, T. Wilberforce, K. Elsaid, E.T. Sayed, M. Ramadan, S.A. Rahman, M.A. Abdelkareem, Recent progress on Carbon-based nanomaterial for phase change materials: prospects and challenges, *Therm. Sci. Eng. Prog.* 23 (2021) 100920, <https://doi.org/10.1016/j.tsep.2021.100920>.
- [33] A. Stonehouse, C. Abeykoon, Thermal properties of phase change materials reinforced with multi-dimensional carbon nanomaterials, *Int. J. Heat Mass Tran.* 183 (2022) 122166, <https://doi.org/10.1016/j.ijheatmasstransfer.2021.122166>.
- [34] T. Li, D. Wu, F. He, R. Wang, Experimental investigation on copper foam/hydrated salt composite phase change material for thermal energy storage, *Int. J. Heat Mass Tran.* 115 (2017) 148–157, <https://doi.org/10.1016/j.ijheatmasstransfer.2017.07.056>.
- [35] T. Li, J.-H. Lee, R. Wang, Y.T. Kang, Enhancement of heat transfer for thermal energy storage application using stearic acid nanocomposite with multi-walled carbon nanotubes, *Energy* 55 (2013) 752–761, <https://doi.org/10.1016/j.energy.2013.04.010>.
- [36] L. Jiang, G. Chen, L. Zhao, M. Li, C. Li, R. Zhang, Preparation and low temperature heat storage properties of 1-hexadecylamine/3D graphene aerogel composite phase change materials, *Sol. Energy Mater. Sol. Cells* 251 (2023) 112118, <https://doi.org/10.1016/j.solmat.2022.112118>.
- [37] K. Wang, R. Wen, Scaphium scaphigerum/graphene hybrid aerogel for composite phase change material with high phase change enthalpy and high thermal conductivity for energy storage, *J. Energy Storage* 58 (2023) 106302, <https://doi.org/10.1016/j.est.2022.106302>.
- [38] Y. Shang, D. Zhang, Preparation and thermal properties of graphene oxide-microencapsulated phase change materials, *Nanoscale Microscale Thermophys. Eng.* 20 (3–4) (2016) 147–157, <https://doi.org/10.1080/15567265.2016.1236865>.
- [39] J.-F. Su, X.-Y. Wang, S. Han, X.-L. Zhang, Y.-D. Guo, Y.-Y. Wang, Y.-Q. Tan, N.-X. Han, W. Li, Preparation and physicochemical properties of microcapsules containing phase-change material with graphene/organic hybrid structure shells, *J. Mater. Chem. A* 5 (45) (2017) 23937–23951, <https://doi.org/10.1039/C7TA06980D>.
- [40] K.S. Novoselov, A.K. Geim, S.V. Morozov, D.-e. Jiang, Y. Zhang, S.V. Dubonos, I.V. Grigorieva, A.A. Firsov, Electric field effect in atomically thin carbon films, *Science* 306 (5696) (2004) 666–669, <https://doi.org/10.1126/science.1102896>.
- [41] L. Bai, C. Hou, X. Wen, J. Guan, Catalysis of oxygen reduction reaction on atomically dispersed copper- and nitrogen-codoped graphene, *ACS Appl. Energy Mater.* 2 (7) (2019) 4755–4762, <https://doi.org/10.1021/acsaem.9b00386>.
- [42] Y. Chen, Y.Y. Yue, S.R. Wang, N. Zhang, J. Feng, H.B. Sun, Graphene as a transparent and conductive electrode for organic optoelectronic devices, *Adv. Electron. Mater.* 5 (10) (2019) 1900247, <https://doi.org/10.1002/aelm.201900247>.
- [43] R.G. Bai, N. Ninan, K. Muthoosamy, S. Manickam, Graphene: a versatile platform for nanotheranostics and tissue engineering, *Rev. Mater. Sci.* 91 (2018) 24–69, <https://doi.org/10.1016/j.pmatsci.2017.08.004>.
- [44] H. Zhang, H. Zhang, A. Aldalabi, X. Zuo, C. Fan, X. Mi, Fluorescent biosensors enabled by graphene and graphene oxide, *Biosens. Bioelectron.* 89 (2017) 96–106, <https://doi.org/10.1016/j.bios.2016.07.030>.
- [45] M.T. Alshamkhani, L.K. Teong, L.K. Putri, A.R. Mohamed, P. Lahijani, M. Mohammadi, Effect of graphite exfoliation routes on the properties of exfoliated graphene and its photocatalytic applications, *J. Environ. Chem. Eng.* 9 (6) (2021) 106506, <https://doi.org/10.1016/j.jece.2021.106506>.
- [46] V. Agarwal, P.B. Zetterlund, Strategies for reduction of graphene oxide—A comprehensive review, *Chem. Eng. J.* 405 (2021) 127018, <https://doi.org/10.1016/j.cej.2020.127018>.
- [47] M. Bahri, S.H. Gebre, M.A. Elagueh, F.T. Dajan, M.G. Sendeku, C. Tlili, D. Wang, Recent advances in chemical vapour deposition techniques for graphene-based nanoarchitectures: from synthesis to contemporary applications, *Coord. Chem. Rev.* 475 (2023) 214910, <https://doi.org/10.1016/j.ccr.2022.214910>.
- [48] S. Wu, Q. Zhang, H. Yang, Y. Ma, T. Zhang, L. Liu, H.-J. Gao, Y. Wang, Advances in two-dimensional heterostructures by mono-element intercalation underneath epitaxial graphene, *Prog. Surf. Sci.* 96 (3) (2021) 100637, <https://doi.org/10.1016/j.progsurf.2021.100637>.
- [49] E. Pop, V. Varshney, A.K. Roy, Thermal properties of graphene: fundamentals and applications, *MRS Bull.* 37 (12) (2012) 1273–1281, <https://doi.org/10.1557/mrs.2012.203>.
- [50] T. Sreethawong, K.W. Shah, S.-Y. Zhang, E. Ye, S.H. Lim, U. Maheswaran, W.Y. Mao, M.-Y. Han, Optimized production of copper nanostructures with high yields for efficient use as thermal conductivity-enhancing PCM dopant, *J. Mater. Chem. A* 2 (10) (2014) 3417–3423, <https://doi.org/10.1039/C3TA14550F>.
- [51] C. Zhu, F. Ran, G. Fang, Thermal properties improvement of lauric acid/iron foam composites with graphene nanoplates as thermal energy storage materials, *J. Energy Storage* 27 (2020) 101163, <https://doi.org/10.1016/j.est.2019.101163>.
- [52] Y. Chen, H. Zhang, C. Xu, R. Cong, G. Fang, Thermal properties of 1-hexadecanol/high density polyethylene/graphene nanoplates composites as form-stable heat storage materials, *Sol. Energy Mater. Sol. Cells* 237 (2022) 111580, <https://doi.org/10.1016/j.solmat.2022.111580>.
- [53] J.I. Prado, L. Lugo, Enhancing the thermal performance of a stearate phase change material with graphene nanoplatelets and MgO nanoparticles, *ACS Appl. Mater. Interfaces* 12 (35) (2020) 39108–39117, <https://doi.org/10.1021/acsaami.0c09643>.
- [54] M. Amin, N. Putra, E.A. Kosasih, E. Prawiro, R.A. Luento, T. Mahlia, Thermal properties of beeswax/graphene phase change material as energy storage for building applications, *Appl. Therm. Eng.* 112 (2017) 273–280, <https://doi.org/10.1016/j.applthermaleng.2016.10.085>.
- [55] X. Xiao, H. Jia, S. Pervaiz, D. Wen, Molten salt/metal foam/graphene nanoparticle phase change composites for thermal energy storage, *ACS Appl. Nano Mater.* 3 (6) (2020) 5240–5251, <https://doi.org/10.1021/acsaanm.0c00648>.
- [56] N. Sheng, R. Zhu, T. Nomura, Z. Rao, C. Zhu, Y. Aoki, H. Habazaki, T. Akiyama, Anisotropically enhanced heat transfer properties of phase change material reinforced by graphene-wrapped carbon fibers, *Sol. Energy Mater. Sol. Cells* 206 (2020) 110280, <https://doi.org/10.1016/j.solmat.2019.110280>.
- [57] A. Chouhan, H.P. Mungse, O.P. Khatri, Surface chemistry of graphene and graphene oxide: a versatile route for their dispersion and tribological applications, *Adv. Colloid Interface Sci.* 283 (2020) 102215, <https://doi.org/10.1016/j.cis.2020.102215>.
- [58] R. Ikram, B.M. Jan, W. Ahmad, An overview of industrial scalable production of graphene oxide and analytical approaches for synthesis and characterization, *J. Mater. Sci. Technol.* 9 (5) (2020) 11587–11610, <https://doi.org/10.1016/j.jmrt.2020.08.050>.
- [59] B.P. Upoma, F. Mahnaz, W.R. Sajal, N. Zahan, M.S.H. Firoz, M.S. Azam, Bio-inspired immobilization of silver and gold on magnetic graphene oxide for rapid catalysis and recyclability, *J. Environ. Chem. Eng.* 8 (3) (2020) 103739, <https://doi.org/10.1016/j.jece.2020.103739>.
- [60] Y. Tian, Z. Yu, L. Cao, X.L. Zhang, C. Sun, D.-W. Wang, Graphene oxide: an emerging electromaterial for energy storage and conversion, *J. Energy Chem.* 55 (2021) 323–344, <https://doi.org/10.1016/j.jechem.2020.07.006>.
- [61] S.Y. Tee, E. Ye, C.P. Teng, Y. Tanaka, K.Y. Tang, K.Y. Win, M.-Y. Han, Advances in photothermal nanomaterials for biomedical, environmental and energy applications, *Nanoscale* 13 (34) (2021) 14268–14286, <https://doi.org/10.1039/D1NR04197E>.
- [62] Y.V. Stebunov, O.A. Aftenieva, A.V. Arsenin, V.S. Volkov, Highly sensitive and selective sensor chips with graphene-oxide linking layer, *ACS Appl. Mater. Interfaces* 7 (39) (2015) 21727–21734, <https://doi.org/10.1021/acsaami.5b04427>.
- [63] K. Huang, J. Li, X. Luan, L. Liu, Z. Yang, C. Wang, Effect of graphene oxide on phase change materials based on disodium hydrogen phosphate dodecahydrate for thermal storage, *ACS Omega* 5 (25) (2020) 15210–15217, <https://doi.org/10.1021/acsomega.0c01184>.
- [64] R. Cao, H. Liu, S. Chen, D. Pei, J. Miao, X. Zhang, Fabrication and properties of graphene oxide-grafted-poly (hexadecyl acrylate) as a solid-solid phase change material, *Compos. Sci. Technol.* 149 (2017) 262–268, <https://doi.org/10.1016/j.compscitech.2017.06.019>.
- [65] W.T. Neo, Q. Ye, M.H. Chua, Q. Zhu, J. Xu, Solution-processable copolymers based on triphenylamine and 3, 4-ethylenedioxythiophene: facile synthesis and multielectrochromism, *Macromol. Rapid Commun.* 41 (21) (2020) 2000156, <https://doi.org/10.1002/marc.202000156>.
- [66] T.P. Vo, M.H. Chua, S.J. Ang, K.L.O. Chin, X.Y.D. Soo, Z.M. Png, T.L.D. Tam, Q. Zhu, D.J. Procter, J. Xu, Pyrrole [3, 4-c] pyridine-1, 2-dione: a new electron acceptor for electrochromic conjugated polymers, *Polym. Chem.* 13 (47) (2022) 6512–6524, <https://doi.org/10.1039/D2PY00863G>.
- [67] M.H. Chua, S.H.G. Toh, P.J. Ong, Z.M. Png, Q. Zhu, S. Xiong, J. Xu, Towards modulating the colour hues of isoindigo-based electrochromic polymers through variation of thiophene-based donor groups, *Polym. Chem.* 13 (7) (2022) 967–981, <https://doi.org/10.1039/D1PY01531A>.
- [68] G. Barouti, S.S. Liow, Q. Dou, H. Ye, C. Orione, S.M. Guillaume, X.J. Loh, New linear and star-shaped thermogelling poly [(R)-3-hydroxybutyrate] copolymers, *Chem. Eur. J.* 22 (30) (2016) 10501–10512, <https://doi.org/10.1002/chem.201601404>.
- [69] Z.M. Png, C.-G. Wang, J. Yeo, J.J.C. Lee, N.E.B. Suratman, Y.L. Tan, H. Liu, P. Wang, M.B.H. Tan, J. Xu, Stimuli-responsive structure-property switchable polymer materials, *Mol. Syst. Des. Eng.* (2023), <https://doi.org/10.1039/D3ME00002H>.
- [70] X. Han, S.-X. Wang, F. Zhong, Y. Lu, Formation of functionalized cyclopentenes via catalytic asymmetric [3+ 2] cycloaddition of acrylamides with an allenolate promoted by dipeptide-derived phosphines, *Synthesis* 2011 (12) (2011) 1859–1864, <https://doi.org/10.1055/s-0030-1260460>.
- [71] Q. Zhu, Y. Lu, Facile synthesis of bicyclic amidines and imidazolines from 1, 2-diamines, *Org. Lett.* 12 (18) (2010) 4156–4159, <https://doi.org/10.1021/ol101747n>.
- [72] H. Zhu, J.M.J. Qiang, C.G. Wang, C.Y. Chan, Q. Zhu, E. Ye, Z. Li, X.J. Loh, Flexible polymeric patch based nanotherapeutics against non-cancer therapy, *Bioact. Mater.* 18 (2022) 471–491, <https://doi.org/10.1016/j.bioactmat.2022.03.034>.
- [73] Q. Zhu, E. Yildirim, X. Wang, A.K.K. Kyaw, T. Tang, X.Y.D. Soo, Z.M. Wong, G. Wu, S.-W. Yang, J. Xu, Effect of substituents in sulfoxides on the enhancement of thermoelectric properties of PEDOT: PSS: experimental and modelling evidence, *Mol. Syst. Des. Eng.* 5 (5) (2020) 976–984, <https://doi.org/10.1039/D0ME00032A>.
- [74] V.P.N. Nguyen, N. Kuo, X.J. Loh, New biocompatible thermogelling copolymers containing ethylene-butylene segments exhibiting very low gelation concentrations, *Soft Matter* 7 (5) (2011) 2150–2159, <https://doi.org/10.1039/C0SM00764A>.
- [75] Z. Li, P.L. Chee, C. Ow, R. Lakshminarayanan, X.J. Loh, Safe and efficient membrane permeabilizing polymers based on PLLA for antibacterial applications, *RSC Adv.* 6 (34) (2016) 28947–28955, <https://doi.org/10.1039/C6RA04531F>.
- [76] X.J. Loh, B.J.H. Yee, F.S. Chia, Sustained delivery of paclitaxel using thermogelling poly (PEG/PPG/PCL urethane) s for enhanced toxicity against cancer cells, *J. Biomed. Mater. Res.* 100 (10) (2012) 2686–2694, <https://doi.org/10.1002/jbm.b.34198>.
- [77] X.J. Loh, S.H. Goh, J. Li, Biodegradable thermogelling poly [(R)-3-hydroxybutyrate]-based block copolymers: micellization, gelation, and

- cytotoxicity and cell culture studies, *J. Phys. Chem. B* 113 (35) (2009) 11822–11830, <https://doi.org/10.1021/jp903984r>.
- [78] X.J. Loh, W. Guerin, S.M. Guillaume, Sustained delivery of doxorubicin from thermogelling poly (PEG/PPG/PTMC urethane)s for effective eradication of cancer cells, *J. Mater. Chem.* 22 (39) (2012) 21249–21256, <https://doi.org/10.1039/C2JM33777K>.
- [79] Y. Xia, H. Zhang, P. Huang, C. Huang, F. Xu, Y. Zou, H. Chu, E. Yan, L. Sun, Graphene-oxide-induced lamellar structures used to fabricate novel composite solid-solid phase change materials for thermal energy storage, *Chem. Eng. J.* 362 (2019) 909–920, <https://doi.org/10.1016/j.cej.2019.01.097>.
- [80] B. Li, T. Liu, L. Hu, Y. Wang, S. Nie, Facile preparation and adjustable thermal property of stearic acid-graphene oxide composite as shape-stabilized phase change material, *Chem. Eng. J.* 215 (2013) 819–826, <https://doi.org/10.1016/j.cej.2012.11.077>.
- [81] S. Park, J. An, J.R. Potts, A. Velamakanni, S. Murali, R.S. Ruoff, Hydrazine-reduction of graphite and graphene oxide, *Carbon* 49 (9) (2011) 3019–3023, <https://doi.org/10.1016/j.carbon.2011.02.071>.
- [82] A.B. Alayande, H.D. Park, J.S. Vrouwenvelder, I.S. Kim, Implications of chemical reduction using hydriodic acid on the antimicrobial properties of graphene oxide and reduced graphene oxide membranes, *Small* 15 (28) (2019) 1901023, <https://doi.org/10.1002/smll.201901023>.
- [83] M. Ceniceros-Reyes, K. Marín-Hernández, U. Sierra, E. Saucedo-Salazar, R. Mendoza-Resendez, C. Luna, P. Hernández-Belmares, O. Rodríguez-Fernández, S. Fernández-Tavizón, E. Hernández-Hernández, Reduction of graphene oxide by in-situ heating experiments in the transmission electron microscope, *Surface. Interfac.* 35 (2022) 102448, <https://doi.org/10.1016/j.surfin.2022.102448>.
- [84] J. Zhou, M. Fang, K. Yang, K. Lu, H. Fei, P. Mu, R. He, MIL-101 (Cr)-NH₂/reduced graphene oxide composite carrier enhanced thermal conductivity and stability of shape-stabilized phase change materials for thermal energy management, *J. Energy Storage* 52 (2022) 104827, <https://doi.org/10.1016/j.est.2022.104827>.
- [85] H.-y. Wu, S.-t. Li, Y.-w. Shao, X.-z. Jin, X.-d. Qi, J.-h. Yang, Z.-w. Zhou, Y. Wang, Melamine foam/reduced graphene oxide supported form-stable phase change materials with simultaneous shape memory property and light-to-thermal energy storage capability, *Chem. Eng. J.* 379 (2020) 122373, <https://doi.org/10.1016/j.cej.2019.122373>.
- [86] C. Cárdenas-Ramírez, F. Jaramillo, M. Gómez, Systematic review of encapsulation and shape-stabilization of phase change materials, *J. Energy Storage* 30 (2020) 101495, <https://doi.org/10.1016/j.est.2020.101495>.
- [87] J.-C. Lee, S. Sugiarto, P.J. Ong, X.Y.D. Soo, X. Ni, P. Luo, Y.Y.K. Hnin, J.S.Y. See, F. Wei, R. Zheng, Lignin-g-polycaprolactone as a form-stable phase change material for thermal energy storage application, *J. Energy Storage* 56 (2022) 106118, <https://doi.org/10.1016/j.est.2022.106118>.
- [88] R. Yeo, W.-Y. Wu, N. Tomczak, R. Ji, S. Wang, X. Wang, J. Kong, H. Liu, K. Goh, J. Xu, Tailoring surface reflectance through nanostructured materials design for energy-efficient applications, *Mater. Today Chem.* 30 (2023) 101593, <https://doi.org/10.1016/j.mtchem.2023.101593>.
- [89] B. Mu, M. Li, Fabrication and characterization of polyurethane-grafted reduced graphene oxide as solid-solid phase change materials for solar energy conversion and storage, *Sol. Energy* 188 (2019) 230–238, <https://doi.org/10.1016/j.solener.2019.05.082>.
- [90] Y. Zhou, X. Liu, D. Sheng, C. Lin, F. Ji, L. Dong, S. Xu, H. Wu, Y. Yang, Polyurethane-based solid-solid phase change materials with in situ reduced graphene oxide for light-thermal energy conversion and storage, *Chem. Eng. J.* 338 (2018) 117–125, <https://doi.org/10.1016/j.cej.2018.01.021>.
- [91] S. Wang, Z. Liang, L. Liu, Y. Cao, Y. Cheng, D. Chen, Preparation of chemically functionalized graphene with excellent dispersibility and tribological properties as lubricant additives by microwave-assisted ball milling, *J. Mol. Liq.* 344 (2021) 117929, <https://doi.org/10.1016/j.molliq.2021.117929>.
- [92] R.K. Layek, A.K. Nandi, A review on synthesis and properties of polymer functionalized graphene, *Polymer* 54 (19) (2013) 5087–5103, <https://doi.org/10.1016/j.polymer.2013.06.027>.
- [93] O. Zabihi, M. Ahmadi, Q. Li, S.M. Fakhroshini, Z.K. Nia, M. Arjmand, K. Parvez, M. Naebe, Simultaneous electrochemical-assisted exfoliation and in situ surface functionalization towards large-scale production of few-layer graphene, *FlatChem* 18 (2019) 100132, <https://doi.org/10.1016/j.flatc.2019.100132>.
- [94] R. Prabhakaran, J. Prasanna Naveen Kumar, D. Mohan Lal, C. Selvam, S. Harish, Constrained melting of graphene-based phase change nanocomposites inside a sphere, *J. Therm. Anal. Calorim.* 139 (2020) 941–952, <https://doi.org/10.1007/s10973-019-08458-4>.
- [95] V. Selvaraj, H. Krishnan, Acidic functionalized graphene dispersed polyethylene glycol nano-phase change material for the active cooling of a simulated heat-generating electronic system, *J. Energy Storage* 45 (2022) 103774, <https://doi.org/10.1016/j.est.2021.103774>.
- [96] L. Tao, S. Chen, H. Liu, N. Han, X. Zhang, Fabrication and characterization of poly (n-alkyl acrylic) ester shape-stable phase-change materials based on UV curing, *ACS Appl. Energy Mater.* 4 (4) (2021) 3358–3368, <https://doi.org/10.1021/acsaem.0c03105>.
- [97] J. Cao, Y. Sim, X.Y. Tan, J. Zheng, S.W. Chien, N. Jia, K. Chen, Y.B. Tay, J.F. Dong, L. Yang, Upcycling silicon photovoltaic waste into thermoelectrics, *Adv. Mater.* 34 (19) (2022) 2110518, <https://doi.org/10.1002/adma.202110518>.
- [98] S.S.F. Duran, D. Zhang, W.Y.S. Lim, J. Cao, H. Liu, Q. Zhu, C.K.I. Tan, J. Xu, X.J. Loh, A. Suwardi, Potential of recycled silicon and silicon-based thermoelectrics for power generation, *Crystals* 12 (3) (2022) 307, <https://doi.org/10.3390/cryst12030307>.
- [99] S. Sugiarto, R. Pong, Y. Tan, Y. Leow, T. Sathasivam, Q. Zhu, X. Loh, D. Kai, Advances in sustainable polymeric materials from lignocellulosic biomass, *Mater. Today Chem.* 26 (2022) 101022, <https://doi.org/10.1016/j.mtchem.2022.101022>.
- [100] J. Zheng, S.F.D. Solco, C.J.E. Wong, S.A. Sia, X.Y. Tan, J. Cao, J.C.C. Yeo, W. Yan, Q. Zhu, Q. Yan, Integrating recyclable polymers into thermoelectric devices for green electronics, *J. Mater. Chem. A* 10 (37) (2022) 19787–19796, <https://doi.org/10.1039/D2TA00386D>.
- [101] J. Cao, J. Zheng, H. Liu, C.K.I. Tan, X. Wang, W. Wang, Q. Zhu, Z. Li, G. Zhang, J. Wu, Flexible elemental thermoelectrics with ultra-high power density, *Mater. Today Energy* 25 (2022) 100964, <https://doi.org/10.1016/j.mtener.2022.100964>.
- [102] X. Fan, B.H. Tan, Z. Li, X.J. Loh, Control of PLA stereoisomers-based polyurethane elastomers as highly efficient shape memory materials, *ACS Sustain. Chem. Eng.* 5 (1) (2017) 1217–1227, <https://doi.org/10.1021/acssuschemeng.6b02652>.
- [103] J.K. Muiruri, J.C.C. Yeo, Q. Zhu, E. Ye, X.J. Loh, Z. Li, Poly (hydroxyalkanoates): production, applications and end-of-life strategies–life cycle assessment nexus, *ACS Sustain. Chem. Eng.* 10 (11) (2022) 3387–3406, <https://doi.org/10.1021/acssuschemeng.1c08631>.
- [104] R. Liang, X. Yang, P.Y.M. Yew, S. Sugiarto, Q. Zhu, J. Zhao, X.J. Loh, L. Zheng, D. Kai, PLA-lignin nanofibers as antioxidant biomaterials for cartilage regeneration and osteoarthritis treatment, *J. Nanobiotechnol.* 20 (1) (2022) 327, <https://doi.org/10.1186/s12951-022-01534-2>.
- [105] S. Chavan, V. Guntapure, Numerical and experimental analysis on thermal energy storage of polyethylene/functionalized graphene composite phase change materials, *J. Energy Storage* 27 (2020) 101045, <https://doi.org/10.1016/j.est.2019.101045>.
- [106] M. Li, C. Wang, AAZO-Graphene composite phase change materials with Photo-thermal conversion performance, *Sol. Energy* 223 (2021) 11–18, <https://doi.org/10.1016/j.solener.2021.05.036>.
- [107] F. Wang, P. Zhang, Y. Mou, M. Kang, M. Liu, L. Song, A. Lu, J. Rong, Synthesis of the polyethylene glycol solid-solid phase change materials with a functionalized graphene oxide for thermal energy storage, *Polym. Test.* 63 (2017) 494–504, <https://doi.org/10.1016/j.polymertesting.2017.09.005>.
- [108] S. Li, L. Kong, H. Wang, H. Xu, J. Li, H. Shi, Thermal performance and shape-stabilization of comb-like polymeric phase change materials enhanced by octadecylamine-functionalized graphene oxide, *Energy Convers. Manag.* 168 (2018) 119–127, <https://doi.org/10.1016/j.enconman.2018.05.014>.
- [109] J. Ge, Y. Wang, H. Wang, H. Mao, J. Li, H. Shi, Thermal properties and shape stabilization of epoxidized methoxy polyethylene glycol composite PCMs tailored by polydopamine-functionalized graphene oxide, *Sol. Energy Mater. Sol. Cells* 208 (2020) 110388, <https://doi.org/10.1016/j.solmat.2019.110388>.
- [110] A.R. Akhiani, M. Mehrali, S. Tahan Latibari, M. Mehrali, T.M.I. Mahlia, E. Sadeghinezhad, H.S.C. Metselaar, One-step preparation of form-stable phase change material through self-assembly of fatty acid and graphene, *J. Phys. Chem. C* 119 (40) (2015) 22787–22796, <https://doi.org/10.1021/acs.jpcc.5b06089>.
- [111] S. Wang, W.Y. Wu, J.C.C. Yeo, X.Y.D. Soo, W. Thitsartarn, S. Liu, B.H. Tan, A. Suwardi, Z. Li, Q. Zhu, Responsive hydrogel dressings for intelligent wound management, *BME Mat* (2023) e12021, <https://doi.org/10.1002/bmm2.12021>.
- [112] M. Liu, Y. Chen, Q. Zhu, J. Tao, C. Tang, H. Ruan, Y. Wu, X.J. Loh, Antioxidant thermogelling formulation for burn wound healing, *Chem. Asian J.* 17 (16) (2022) e202200396, <https://doi.org/10.1002/asia.202200396>.
- [113] S. Sun, Q. Yan, M. Wu, X. Zhao, Carbon aerogel based materials for secondary batteries, *Sustain. Mater. Technol.* 30 (2021) e00342, <https://doi.org/10.1016/j.susmat.2021.e00342>.
- [114] C. Shi, J. Huang, Y. Tang, Z. Cen, Z. Wang, S. Liu, R. Fu, A hierarchical porous carbon aerogel embedded with small-sized TiO₂ nanoparticles for high-performance Li-S batteries, *Carbon* 202 (2023) 59–65, <https://doi.org/10.1016/j.carbon.2022.09.086>.
- [115] C. Zhang, D. Liao, Y. Wang, P. Xie, M. Li, L. Zhou, Y. Chen, H. Liu, Pickering emulsion strategy for high compressive carbon aerogel for use as insulation material and dual-mode pressure/magnetic sensor, *Ceram. Int.* 49 (5) (2023) 8121–8131, <https://doi.org/10.1016/j.ceramint.2022.10.335>.
- [116] L. Cao, C. Wang, Y. Huang, Structure optimization of graphene aerogel-based composites and applications in batteries and supercapacitors, *Chem. Eng. J.* 454 (2023) 140094, <https://doi.org/10.1016/j.cej.2022.140094>.
- [117] W. Wu, M. Du, H. Shi, Q. Zheng, Z. Bai, Application of graphene aerogels in oil spill recovery: a review, *Sci. Total Environ.* 856 (2023) 159107, <https://doi.org/10.1016/j.scitotenv.2022.159107>.
- [118] X. Zhang, J. Zhou, Y. Zheng, H. Wei, Z. Su, Graphene-based hybrid aerogels for energy and environmental applications, *Chem. Eng. J.* 420 (2021) 129700, <https://doi.org/10.1016/j.cej.2021.129700>.
- [119] L. Cao, D. Zhang, Application potential of graphene aerogel in paraffin phase change composites: experimental study and guidance based on numerical simulation, *Sol. Energy Mater. Sol. Cells* 223 (2021) 110949, <https://doi.org/10.1016/j.solmat.2020.110949>.
- [120] J. Huang, B. Zhang, M. He, X. Huang, G. Wu, G. Yin, Y. Cui, Preparation of anisotropic reduced graphene oxide/BN/paraffin composite phase change materials and investigation of their thermal properties, *J. Mater. Sci.* 55 (2020) 7337–7350, <https://doi.org/10.1007/s10853-020-04514-9>.
- [121] X. Zheng, X. Gao, Z. Huang, Z. Li, Y. Fang, Z. Zhang, Form-stable paraffin/graphene aerogel/copper foam composite phase change material for solar energy conversion and storage, *Sol. Energy Mater. Sol. Cells* 226 (2021) 111083, <https://doi.org/10.1016/j.solmat.2021.111083>.
- [122] F. Xue, X.-z. Jin, W.-y. Wang, X.-d. Qi, J.-h. Yang, Y. Wang, Melamine foam and cellulose nanofiber co-mediated assembly of graphene nanoplatelets to construct three-dimensional networks towards advanced phase change materials, *Nanoscale* 12 (6) (2020) 4005–4017, <https://doi.org/10.1039/C9NR10696K>.

- [123] C. Yu, H. Kim, J.R. Youn, Y.S. Song, Enhancement of structural stability of graphene aerogel for thermal energy harvesting, *ACS Appl. Energy Mater.* 4 (10) (2021) 11666–11674, <https://doi.org/10.1021/acsaem.1c02390>.
- [124] D. Wei, C. Wu, G. Jiang, X. Sheng, Y. Xie, Lignin-assisted construction of well-defined 3D graphene aerogel/PEG form-stable phase change composites towards efficient solar thermal energy storage, *Sol. Energy Mater. Sol. Cells* 224 (2021) 111013, <https://doi.org/10.1016/j.solmat.2021.111013>.
- [125] Z. Bao, N. Bing, H.-R. Yao, Y. Zhang, H. Xie, W. Yu, Three-dimensional interpenetrating network phase-change composites with high photothermal conversion and rapid heat storage and release, *ACS Appl. Energy Mater.* 4 (8) (2021) 7710–7720, <https://doi.org/10.1021/acsaem.1c01061>.
- [126] L.-S. Tang, J. Yang, R.-Y. Bao, Z.-Y. Liu, B.-H. Xie, M.-B. Yang, W. Yang, Polyethylene glycol/graphene oxide aerogel shape-stabilized phase change materials for photo-to-thermal energy conversion and storage via tuning the oxidation degree of graphene oxide, *Energy Convers. Manag.* 146 (2017) 253–264, <https://doi.org/10.1016/j.enconman.2017.05.037>.
- [127] S. Xi, M. Wang, L. Wang, H. Xie, W. Yu, 3D reduced graphene oxide aerogel supported TiO₂-x for shape-stable phase change composites with high photothermal efficiency and thermal conductivity, *Sol. Energy Mater. Sol. Cells* 226 (2021) 111068, <https://doi.org/10.1016/j.solmat.2021.111068>.
- [128] Y. Cai, N. Zhang, Y. Yuan, W. Zhong, N. Yu, Multi-energy driven form-stable phase change materials based on SEBS and reduced graphene oxide aerogel, *Sol. Energy Mater. Sol. Cells* 233 (2021) 111390, <https://doi.org/10.1016/j.solmat.2021.111390>.
- [129] P. Chen, X. Gao, Y. Wang, T. Xu, Y. Fang, Z. Zhang, Metal foam embedded in SEBS/paraffin/HDPE form-stable PCMs for thermal energy storage, *Sol. Energy Mater. Sol. Cells* 149 (2016) 60–65, <https://doi.org/10.1016/j.solmat.2015.12.041>.
- [130] Q. Zhang, Y. Zhao, J. Feng, Systematic investigation on shape stability of high-efficiency SEBS/paraffin form-stable phase change materials, *Sol. Energy Mater. Sol. Cells* 118 (2013) 54–60, <https://doi.org/10.1016/j.solmat.2013.07.035>.
- [131] X.J. Loh, V.P.N. Nguyen, N. Kuo, J. Li, Encapsulation of basic fibroblast growth factor in thermogelling copolymers preserves its bioactivity, *J. Mater. Chem.* 21 (7) (2011) 2246–2254, <https://doi.org/10.1039/C0JM03051A>.
- [132] S. Wang, P.J. Ong, S. Liu, W. Thitsartarn, M.J.B.H. Tan, A. Suwardi, Q. Zhu, X.J. Loh, Recent advances in host-guest supramolecular hydrogels for biomedical applications, *Chem. Asian J.* 17 (18) (2022) e202200608, <https://doi.org/10.1002/asia.202200608>.
- [133] H. Lambert, A. Castillo Bonillo, Q. Zhu, Y.-W. Zhang, T.-C. Lee, Supramolecular gating of guest release from cucurbit [7] uril using de novo design, *Npj Comput. Mater.* 8 (1) (2022) 21, <https://doi.org/10.1038/s41524-022-00702-0>.
- [134] E. Ye, H. Tan, S. Li, W.Y. Fan, Self-Organization of spherical, core-shell palladium aggregates by laser-induced and thermal decomposition of [Pd (PPh₃)₄], *Angew. Chem. Int. Ed.* 45 (7) (2006) 1120–1123, <https://doi.org/10.1002/anie.200503408>.
- [135] M. Wu, T. Li, P. Wang, S. Wu, R. Wang, J. Lin, Dual-encapsulated highly conductive and liquid-free phase change composites enabled by polyurethane/graphite nanoplatelets hybrid networks for efficient energy storage and thermal management, *Small* 18 (9) (2022) 2105647, <https://doi.org/10.1002/smll.202105647>.
- [136] J.J.C. Lee, N.J.X. Lim, P. Wang, H. Liu, S. Wang, C.-L.K. Lee, D. Kai, F. Wei, R. Ji, B.H. Tan, Silicon-containing additives in encapsulation of phase change materials for thermal energy storage, *WV Ann. Rev. Funct. Mater.* (2022), <https://doi.org/10.1142/S2810922822300070>.
- [137] X. Zhang, Y. Zhang, H. Li, Z. Chen, Enhanced thermal conductivity and photothermal effect of microencapsulated n-octadecane phase change material with calcium carbonate-polydopamine hierarchical shell for solar energy storage, *Sol. Energy Mater. Sol. Cells* 256 (2023) 112336, <https://doi.org/10.1016/j.solmat.2023.112336>.
- [138] Q. Zhao, F. He, Q. Zhang, J. Fan, R. He, K. Zhang, H. Yan, W. Yang, Microencapsulated phase change materials based on graphene Pickering emulsion for light-to-thermal energy conversion and management, *Sol. Energy Mater. Sol. Cells* 203 (2019) 110204, <https://doi.org/10.1016/j.solmat.2019.110204>.
- [139] C. Wu, D. Hou, B. Yin, S. Li, Synthesis and application of new core-shell structure via Pickering emulsion polymerization stabilized by graphene oxide, *Compos. B Eng.* 247 (2022) 110285, <https://doi.org/10.1016/j.compositesb.2022.110285>.
- [140] B. Chi, Y. Yao, S. Cui, X. Jin, Preparation of graphene oxide coated tetradecanol/expanded graphite composite phase change material for thermal energy storage, *Mater. Lett.* 282 (2021) 128666, <https://doi.org/10.1016/j.matlet.2020.128666>.
- [141] Y. Liang, Z. Tao, Q. Guo, Z. Liu, Sponge gourd-bioinspired phase change material with high thermal conductivity and excellent shape-stability, *J. Energy Storage* 39 (2021) 102634, <https://doi.org/10.1016/j.est.2021.102634>.
- [142] Z. Huang, C. Wang, L. Zhou, C. Wu, Thermal conductivity enhancement and shape stability of phase-change materials using high-strength 3D graphene skeleton, *Surface. Interfac.* 26 (2021) 101338, <https://doi.org/10.1016/j.surfin.2021.101338>.
- [143] Y. Ren, P. Hao, Modification mechanism and enhanced low-temperature performance of asphalt mixtures with graphene-modified phase-change microcapsules, *Construct. Build. Mater.* 320 (2022) 126301, <https://doi.org/10.1016/j.conbuildmat.2021.126301>.
- [144] Z. Jiang, W. Yang, F. He, C. Xie, J. Fan, J. Wu, K. Zhang, Modified phase change microcapsules with calcium carbonate and graphene oxide shells for enhanced energy storage and leakage prevention, *ACS Sustain. Chem. Eng.* 6 (4) (2018) 5182–5191, <https://doi.org/10.1021/acsschemeng.7b04834>.
- [145] L. He, S. Mo, P. Lin, L. Jia, Y. Chen, Z. Cheng, D-mannitol@ silica/graphene oxide nanoencapsulated phase change material with high phase change properties and thermal reliability, *Appl. Energy* 268 (2020) 115020, <https://doi.org/10.1016/j.apenergy.2020.115020>.
- [146] D.-Z. Chen, S.-Y. Qin, G.C. Tsui, C.-y. Tang, X. Ouyang, J.-h. Liu, J.-N. Tang, J.-D. Zhuo, Fabrication, morphology and thermal properties of octadecylamine-grafted graphene oxide-modified phase-change microcapsules for thermal energy storage, *Compos. B Eng.* 157 (2019) 239–247, <https://doi.org/10.1016/j.compositesb.2018.08.066>.
- [147] W. Wang, H. Cao, J. Liu, S. Jia, L. Ma, X. Guo, W. Sun, A thermal energy storage composite by incorporating microencapsulated phase change material into wood, *RSC Adv.* 10 (14) (2020) 8097–8103, <https://doi.org/10.1039/C9RA09549G>.
- [148] A. Khezri, M. Sahebi, M. Mohammadi, Fabrication and Thermal properties of graphene nanoplatelet-enhanced phase change materials based on paraffin encapsulated by melamine-formaldehyde, *J. Therm. Anal. Calorim.* 147 (14) (2022) 7683–7691, <https://doi.org/10.1007/s10973-021-11085-7>.
- [149] H. Wang, Y. Zhang, E. Ci, X. Li, J. Li, An experimental study in full spectra of solar-driven magnesium nitrate hexahydrate/graphene composite phase change materials for solar thermal storage applications, *J. Energy Storage* 38 (2021) 102536, <https://doi.org/10.1016/j.est.2021.102536>.
- [150] H. Cui, P. Wang, H. Yang, T. Xu, Design and preparation of salt hydrate/graphene oxide@ SiO₂/SiC composites for efficient solar thermal utilization, *Sol. Energy Mater. Sol. Cells* 236 (2022) 111524, <https://doi.org/10.1016/j.solmat.2021.111524>.
- [151] X. Lu, B. Liang, X. Sheng, T. Yuan, J. Qu, Enhanced thermal conductivity of polyurethane/wood powder composite phase change materials via incorporating low loading of graphene oxide nanosheets for solar thermal energy storage, *Sol. Energy Mater. Sol. Cells* 208 (2020) 110391, <https://doi.org/10.1016/j.solmat.2019.110391>.
- [152] G. Yang, L. Zhao, C. Shen, Z. Mao, H. Xu, X. Feng, B. Wang, X. Sui, Boron nitride microsheets bridged with reduced graphene oxide as scaffolds for multifunctional shape stabilized phase change materials, *Sol. Energy Mater. Sol. Cells* 209 (2020) 110441, <https://doi.org/10.1016/j.solmat.2020.110441>.
- [153] Z. Tao, X. Chen, M. Yang, X. Xu, Y. Sun, Y. Li, J. Wang, G. Wang, Three-dimensional rGO@ sponge framework/paraffin wax composite shape-stabilized phase change materials for solar-thermal energy conversion and storage, *Sol. Energy Mater. Sol. Cells* 215 (2020) 110600, <https://doi.org/10.1016/j.solmat.2020.110600>.
- [154] Y. Zhang, K. Wang, W. Tao, D. Li, Preparation of microencapsulated phase change materials used graphene oxide to improve thermal stability and its incorporation in gypsum materials, *Construct. Build. Mater.* 224 (2019) 48–56, <https://doi.org/10.1016/j.conbuildmat.2019.06.227>.
- [155] W. Jin, L. Jiang, L. Chen, T. Yin, Y. Gu, M. Guo, X. Yan, X. Ben, Enhancement of thermal conductivity by graphene as additive in lauric-stearic acid/treated diatomite composite phase change materials for heat storage in building envelope, *Energy Build.* 246 (2021) 111087, <https://doi.org/10.1016/j.enbuild.2021.111087>.
- [156] W. Jin, L. Jiang, L. Chen, Y. Gu, M. Guo, L. Han, X. Ben, H. Yuan, Z. Lin, Preparation and characterization of capric-stearic acid/montmorillonite/graphene composite phase change material for thermal energy storage in buildings, *Construct. Build. Mater.* 301 (2021) 124102, <https://doi.org/10.1016/j.conbuildmat.2021.124102>.
- [157] E. Shamsaei, F. Basquiroto de Souza, A. Fouladi, K. Sagoe-Crentsil, W. Duan, Graphene oxide-based mesoporous calcium silicate hydrate sandwich-like structure: synthesis and application for thermal energy storage, *ACS Appl. Energy Mater.* 5 (1) (2022) 958–969, <https://doi.org/10.1021/acsaem.1c03356>.
- [158] B.E. Jia, A.Q. Thang, C. Yan, C. Liu, C. Lv, Q. Zhu, J. Xu, J. Chen, H. Pan, Q. Yan, Rechargeable aqueous aluminum-ion battery: progress and outlook, *Small* 18 (43) (2022) 2107773, <https://doi.org/10.1002/smll.202107773>.
- [159] Y. Yang, S. Bremner, C. Menictas, M. Kay, Battery energy storage system size determination in renewable energy systems: a review, *Renew. Sustain. Energy Rev.* 91 (2018) 109–125, <https://doi.org/10.1016/j.rser.2018.03.047>.
- [160] Y. Shi, C. Lee, X. Tan, L. Yang, Q. Zhu, X. Loh, J. Xu, Q. Yan, Atomic-level metal electrodeposition: synthetic strategies, applications, and catalytic mechanism in electrochemical energy conversion, *Small Struct.* 3 (3) (2022) 2100185, <https://doi.org/10.1002/ssr.202100185>.
- [161] S. Gu, J. Kong, L. Xing, X. Zhu, J. Xu, C. Chen, Z. Zhang, Feasible regeneration of cathode material from spent portable electronics batteries via nano-bubbles enhanced leaching, *J. Clean. Prod.* 368 (2022) 133199, <https://doi.org/10.1016/j.jclepro.2022.133199>.
- [162] N. Ding, X. Wang, Y. Hou, S. Wang, X. Li, D.W.H. Fam, Y. Zong, Z. Liu, Rational design of a high-energy LiNiO₂ 8CoO₂ 15AlO₂ 05O₂ cathode for Li-ion batteries, *Solid State Ionics* 323 (2018) 72–77, <https://doi.org/10.1016/j.ssi.2018.05.020>.
- [163] M. Wang, Y. Wang, C. Zhang, F. Yu, Cascade phase change based on hydrate salt/carbon hybrid aerogel for intelligent battery thermal management, *J. Energy Storage* 45 (2022) 103771, <https://doi.org/10.1016/j.est.2021.103771>.
- [164] Y. Lin, Q. Kang, H. Wei, H. Bao, P. Jiang, Y.-W. Mai, X. Huang, Spider web-inspired graphene skeleton-based high thermal conductivity phase change nanocomposites for battery thermal management, *Nano-Micro Lett.* 13 (1) (2021) 180, <https://doi.org/10.1007/s40820-021-00702-7>.
- [165] P. Goli, S. Legedza, A. Dhar, R. Salgado, J. Renteria, A.A. Balandin, Graphene-enhanced hybrid phase change materials for thermal management of Li-ion batteries, *J. Power Sources* 248 (2014) 37–43, <https://doi.org/10.1016/j.jpowsour.2013.08.135>.
- [166] T. Ruiz-Calleja, M. Bonet-Aracil, J. Gisbert-Payá, E. Bou-Belda, Analysis of the influence of graphene and phase change microcapsules on thermal behavior of cellulosic fabrics, *Mater. Today Commun.* 25 (2020) 101557, <https://doi.org/10.1016/j.mtcomm.2020.101557>.

- [167] X. Lin, S. Jia, J. Liu, X. Li, X. Guo, W. Sun, Thermally induced flexible wood based on phase change materials for thermal energy storage and management, *J. Mater. Sci.* 56 (2021) 16570–16581, <https://doi.org/10.1007/s10853-021-06239-9>.
- [168] B. Nie, J. Chen, Z. Du, Y. Li, T. Zhang, L. Cong, B. Zou, Y. Ding, Thermal performance enhancement of a phase change material (PCM) based portable box for cold chain applications, *J. Energy Storage* 40 (2021) 102707, <https://doi.org/10.1016/j.est.2021.102707>.
- [169] H.M. Ali, Analysis of heat pipe-aided graphene-oxide based nanoparticle-enhanced phase change material heat sink for passive cooling of electronic components, *J. Therm. Anal. Calorim.* 146 (2021) 277–286, <https://doi.org/10.1007/s10973-020-09946-8>.